Jyothi Engineering College

MNRAT309-
ROBOTICS AND SENSORS



UNIT I

An introduction to the origins, history, anatomy, classification, specifications, and applications of industrial robots.

From science fiction to the factory floor

Introduction

The field of robotics has its origins in science fiction. The term “robot” was derived from the English translation of a fantasy play written in Czechoslovakia around 1920. It took another 40 years before the modern technology of industrial robotics began. Today, robots are highly automated mechanical manipulators controlled by computers.

Robotics

An applied engineering science combining machine tool technology and computer science — including machine design, production theory, microelectronics, computer programming, and artificial intelligence.

“The science of designing and building robots suitable for real-life application in automated manufacturing and other non-manufacturing environments.”

Disciplines Combined

- **Machine Design**
- **Production Theory**
- **Microelectronics**
- **Computer Programming**
- **Artificial Intelligence**

ROBOTICS

- Defined by Robotics Industry Association (RIA) as

  - a re-programmable, multifunctional manipulator designed to move material, parts, tools or specialized devices through variable programmed motion for a variety of tasks

- possess certain anthropomorphic characteristics.
  - mechanical arm
  - sensors to respond to input
  - Intelligence to make decisions

1.1 History of Robotics

1920 Czech playwright Karel Capek introduces the word “robot” in his play Rossum’s Universal Robots.

1954 George Devol replaces the slave manipulator in a teleoperator with CNC programmability — the first “industrial robot,” the Programmable Article Transfer Device.

1956 Joseph Engleberger buys the rights to Devol’s robot and founds the Unimation Company; Devol applies for a patent for ‘Unimate,’ the first programmable robot.

1966–68 ‘Shakey,’ a mobile robot developed by SRI, becomes capable of planning, route-finding, and moving objects.

1977–92 The three-wheeled mobile robot Hilaire is developed at LAAS, Toulouse, France — still in use.

1978 Unimation introduces PUMA (Programmable Universal Machine for Assembly).

1.1 History of Robotics

1979 SCARA (Selective Compliant Articulated Robot for Assembly) introduced in Japan and the US by Adept Technologies.

1980s Legged and hopping robots emerge — BIPER (Shimoyama) and Raibert's hoppers (1986).

1984–91 MIT researchers build mobile robots from LEGO blocks (precursor to Mindstorms); Rodney Brooks creates insect robots — the birth of behavioral robotics.

1986 Honda begins work on its first humanoid, 'E0,' which later becomes ASIMO.

1988–99 SCAMP debuts as the first robot pet with emotions; Sony later introduces AIBO, an autonomous robotic dog.

2002–10 iRobot launches the Roomba; Osaka University unveils 'Actroid'; NASA and GM develop the humanoid astronaut Robonaut-2.

Common Types of Industrial Robots

Articulated

Rotary joints from simple two-joint arms up to 10+ joints. Connected to the base by a twisting joint; industrial robots commonly use four or six axes.

Cartesian

Also called rectilinear or gantry robots. Three linear joints along X, Y, Z, with an optional wrist for rotational movement; delivers linear motion along each axis.

Cylindrical

At least one rotary joint at the base plus one prismatic joint connecting links, producing a cylindrical-shaped work envelope.

Polar (Spherical)

Arm joined to the base by a twisting joint with two rotary joints and one linear joint, forming a spherical work envelope.

SCARA

A selectively compliant, primarily cylindrical arm for assembly, with two parallel joints giving compliance in one plane.

Delta

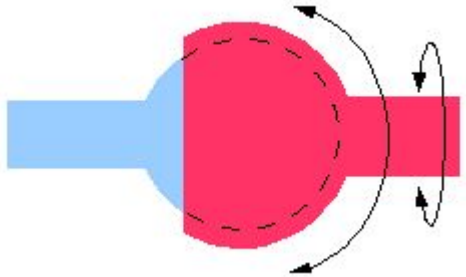
Spider-like robots of jointed parallelograms on a common base, moving a single end-effector in a dome-shaped area — fast and precise for food, pharma, and electronics.

1.3 Robot Components (Anatomy)

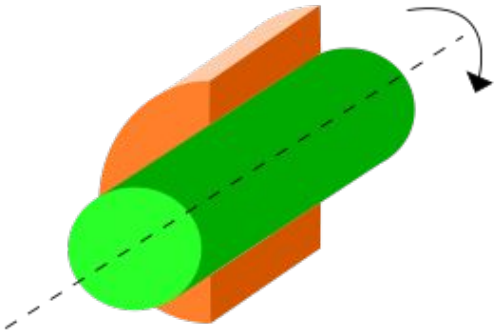
- The body of a robot moves in some 2D or 3D space, by means of applying forces to the robot's own body or its environment. The body can have a wide range of forms, ranging from a ground vehicle with wheels to a complex humanoid robot.
- Most robots are usually made of metal or some other rigid material, where rigid sections known as links are connected by movable joints at their articulation points.

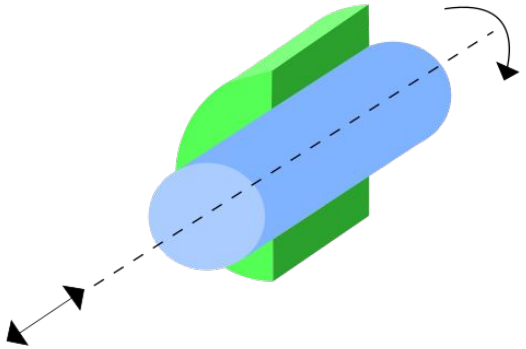
Link. A rigid component of a robot, often attached to other links via joints.

Joint. An articulation point between two robot links.









1.3 Robot Components (Anatomy)

A robot is a system that integrates the following elements into a whole:

Manipulator / Rover

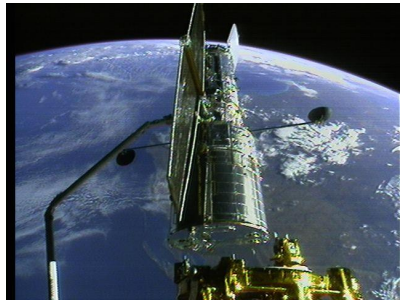
Main body of the robot — links, joints, and structural elements.

End Effector

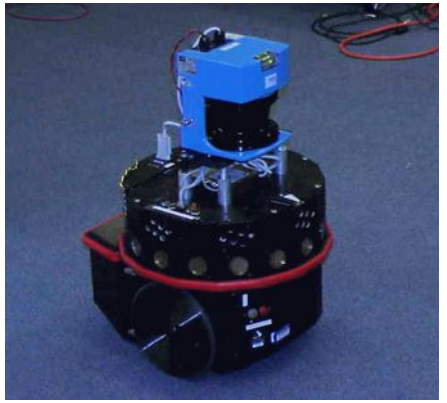
Handles objects, connects to machines, or performs tasks — from a space-shuttle tool to a small gripper.

Actuators

The “muscles” of the manipulator: servomotors, stepper motors, pneumatic cylinders.







1.3 Robot Components (Anatomy)

A robot is a system that integrates the following elements into a whole:

Type	Examples	Purpose
Proprioceptive	joint encoders (relative or absolute), joint limit contact switches	Measure the position of a link relative to another.
Inertial	gyro, accelerometer, IMU	Measure the acceleration, velocity, or position of a link.
Tactile	contact switch, force/torque sensor (load cell), scale, pressure sensor, tactile array	Detect external forces on the robot's body.
Visual	monocular camera, stereo camera, laser rangefinder, depth sensor	Measure light intensity from the robot's surroundings. Some sensors estimate distance from the device to objects in the world via triangulation or time-of-flight.
Other	GPS, sonar, temperature (especially motor temperature), gas sensors, battery voltage	

Sensors

Collect internal state data or communicate with the environment — vision, touch, tactile sensors.

1.3 Robot Components (Anatomy)



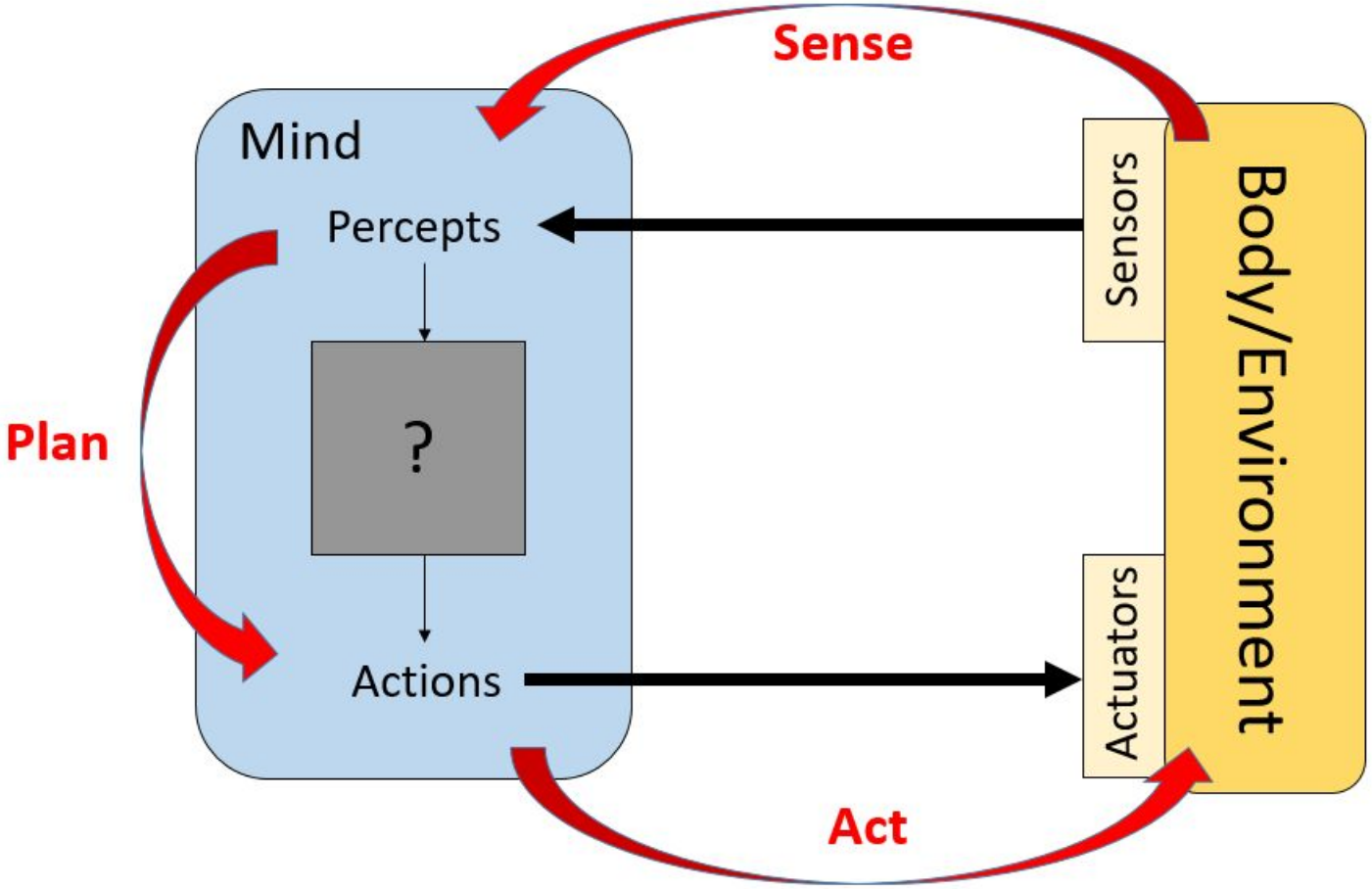
A robot is a system that integrates the following elements into a whole:

Controller

Receives data from the computer, controls actuator motion, and coordinates it with sensory feedback.

Processor

The “brain” of the robot — calculates joint motions and oversees controller and sensor actions.



Three Groups of Software

1 Operating Systems

Operates the computer.

2 Robotic Software

Calculates the necessary motions of each joint and informs the controller. May range from machine language to sophisticated modern robot languages.

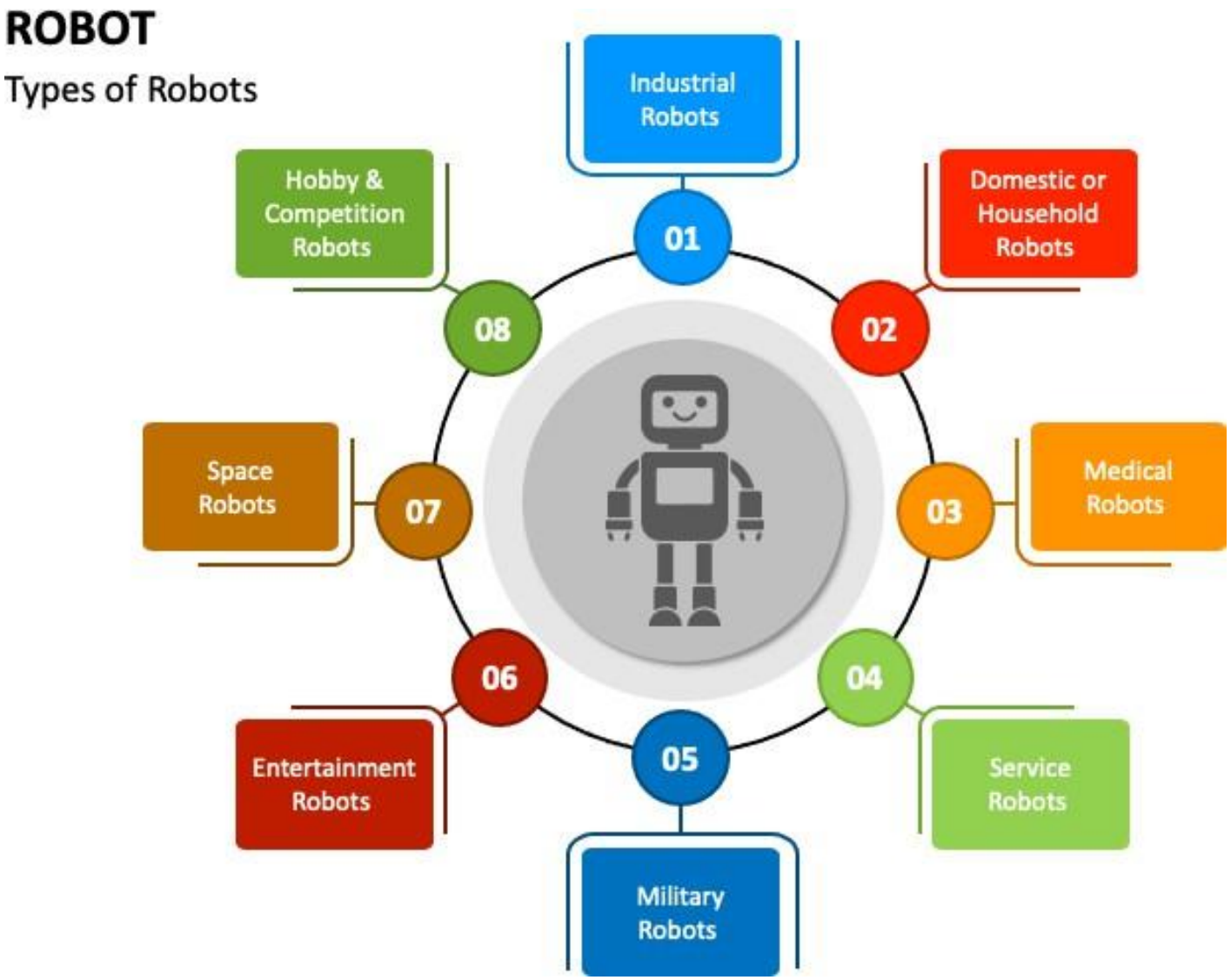
3 Routines & Application Programs

A collection of routines and application programs supporting robot operation.

Joints and Links

- The manipulator of an industrial robot consists of a series of joints and links.
- Robot anatomy is the study of these joints and links and other aspects of physical construction.
- A robotic joint provides relative motion between two links of the robot.
- Each joint provides a certain degree of freedom of motion.
- Mostly, 1 degree of freedom (dof) is associated with each joint.

1.4 Classification of Robots



1.5 Classification by Co-ordinate System

Industrial robots are available in a wide variety of sizes, shapes, and physical configurations. Most commercially available robots use one of four basic configurations:

1. Polar (Spherical)

A telescoping arm raised or lowered about a horizontal pivot, mounted on a rotating base. Gives the arm a spherical work space — hence "spherical coordinate" robot.

2. Cylindrical

A vertical column with a slide that moves up/down and radially with respect to the column, achieving a cylindrical work space.

3. Cartesian (XYZ)

Three perpendicular slides construct the X, Y, and Z axes. Also called an XYZ or rectilinear robot; operates within a rectangular work envelope.

4. Jointed-Arm (Revolute)

Similar to a human arm, with two straight components (forearm and upper arm) on a vertical pedestal, joined by two rotary joints at the shoulder and elbow.

1.5 Classification of Mobile Robot

Industrial robots are available in a wide variety of sizes, shapes, and physical configurations. Most commercially available robots use one of four basic configurations:

1. Wheeled Robots:

Navigate ground surfaces using motorized wheels.

Application: Used in factories and warehouses for material transport (AGV)

2. Aerial Robots (UAVs/Drones):

Aircraft without on-board human pilots, used for surveillance and autonomous flight.

3. Legged Robots:

Use articulated limbs for locomotion. They are more versatile than wheeled robots and can traverse rugged terrains by imitating animals or humans (biomimicry).
Application: Used in rough or uneven terrains such as search and rescue operations

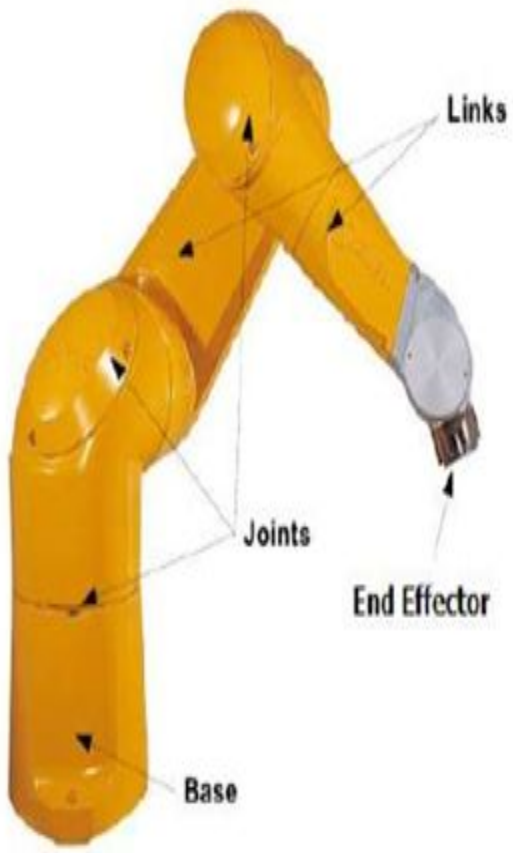
4. Surface Water Robots:

Used specifically for tasks like the removal and collection of surface waste from water bodies

Structure of Robot Manipulators

The robotic manipulators are composed of:

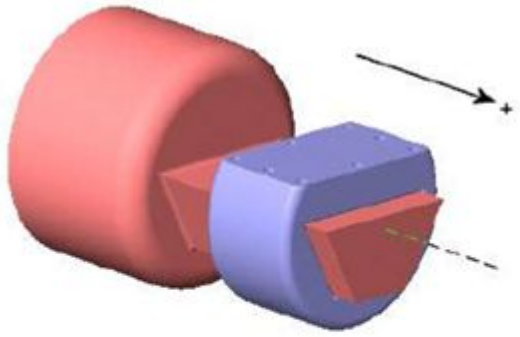
- **Kinematic open chain composed of Rigid Links and Joints.**
- **The BASE: can be either fixed in the work environment or placed on a mobile platform.**
- **End-Effector: Tool is located at the end, used to execute the desired operations [gripper or specific tool].**



Each joint connects two links together.

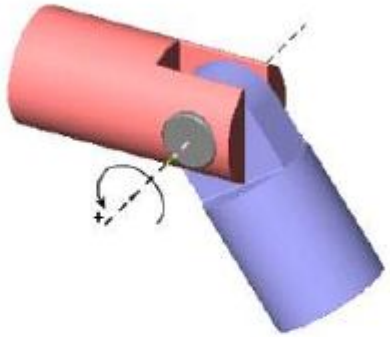
Structure of Robot Manipulators (Types of Joints)

Linear (Prismatic) Joint



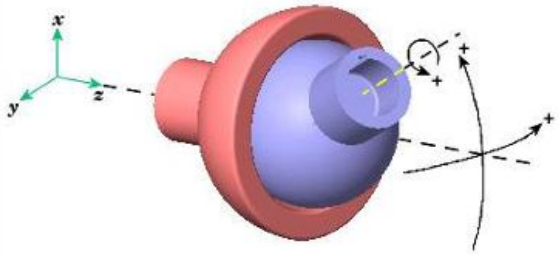
- Allows **translation** between two links.
- It is represented by symbol P.
- The joint variable is displacement d.

Rotary (Revolute) Joint



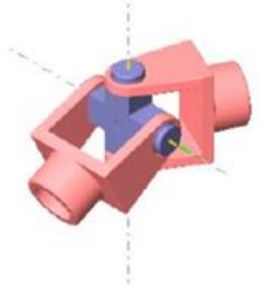
- Allows **rotation** between two links.
- It is represented by symbol R.
- The joint variable is angle θ.

Spherical Joint



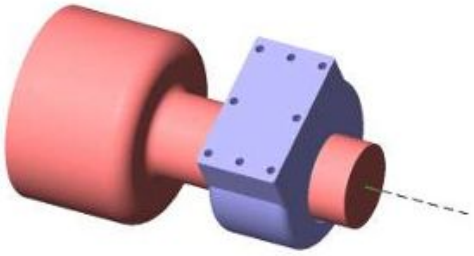
- Allows rotation around three axes.
- It is represented by symbol S.
- The joint variables are θ, γ and ψ.

Universal Joint



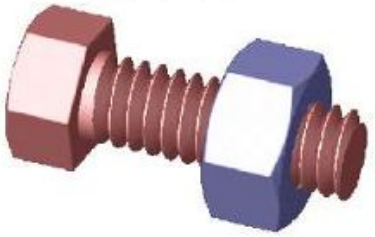
- Allows rotation around two axes.
- It is represented by symbol U.
- The joint variables are θ_1 and θ_2.

Cylindrical Joint



- Allows rotation and translation.
- It is represented by symbol C.

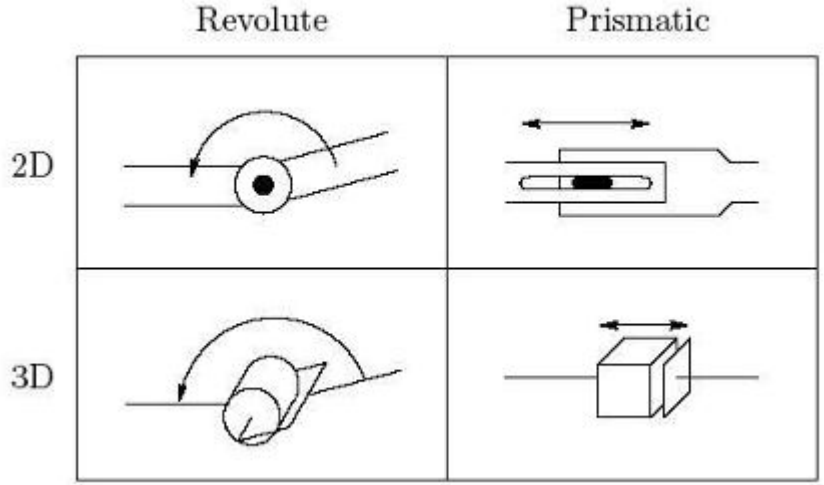
Screw Joint

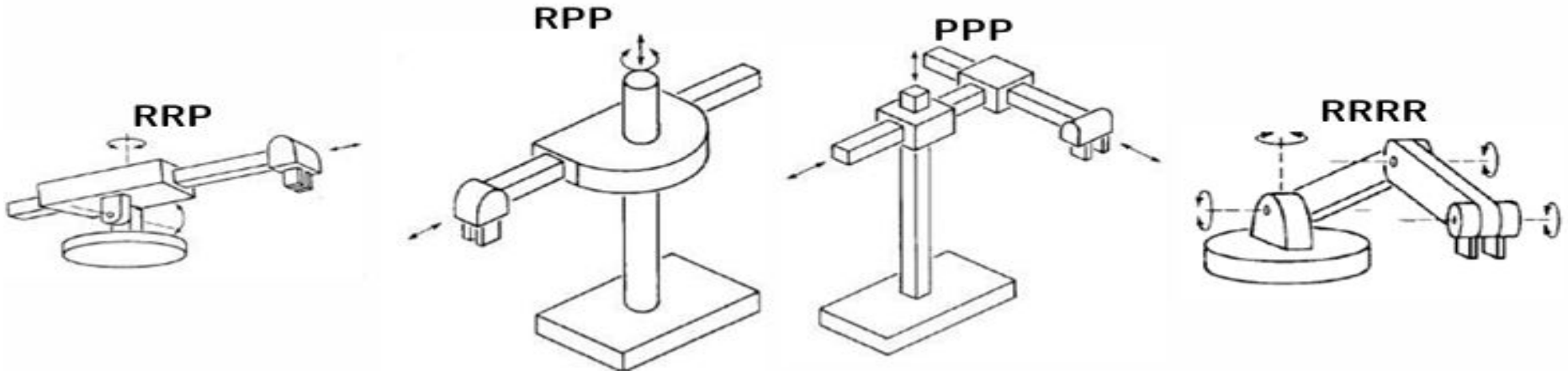


- Allows rotation and a constrained translation.
- It is represented by symbol SC.

Structure of Robot Manipulators

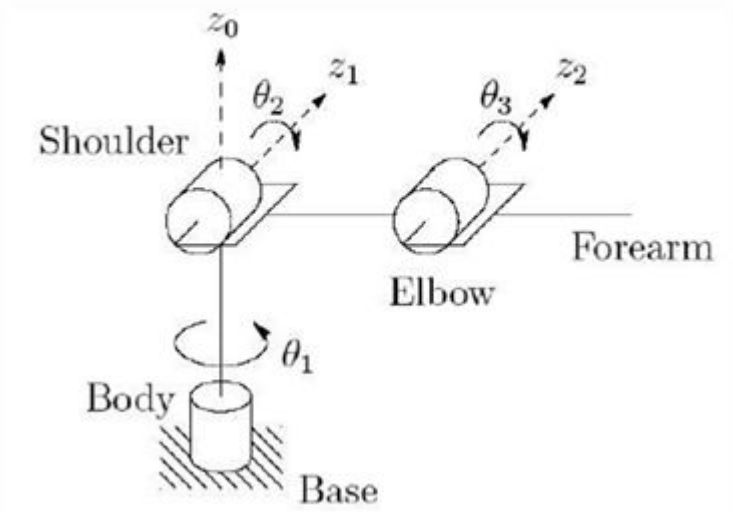
The two common joints in serial robot manipulators are (Prismatic and Revolute) joints.



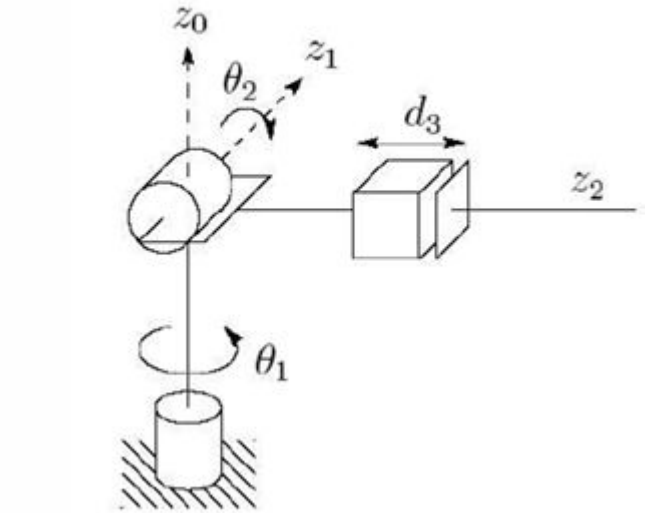


Common Kinematic Arrangements:

[1] Articulated Manipulator (RRR):

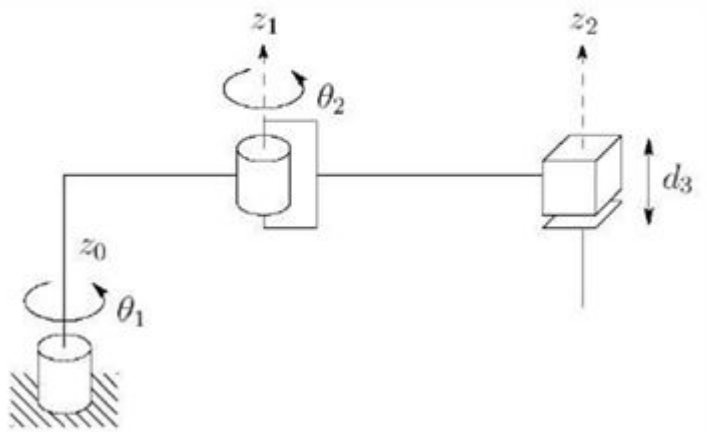


[2] Spherical Manipulator (RRP):



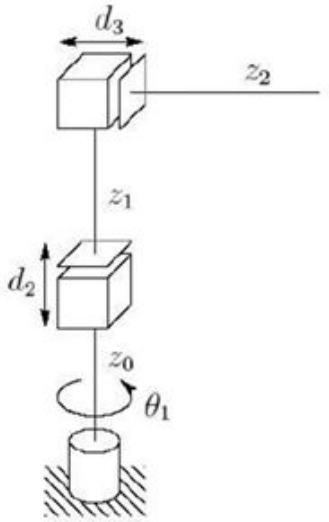
[3] SCARA Manipulator (RRP):

Selective **C**ompliant **A**rticulated **R**obot for **A**ssembly:

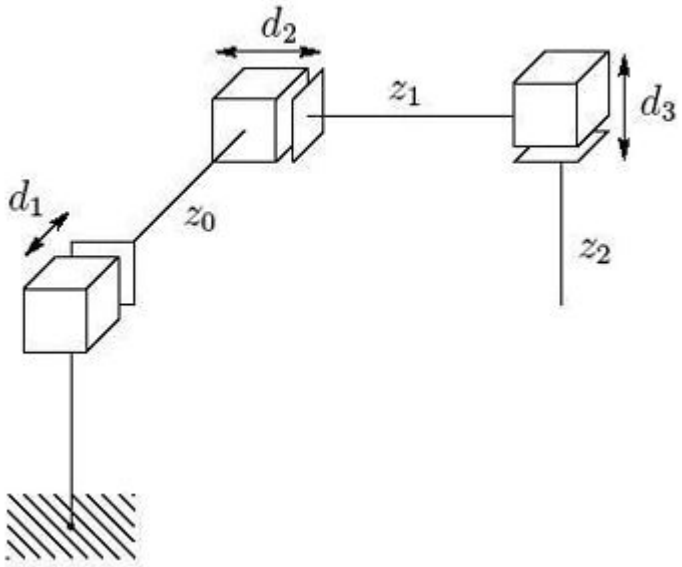


Common Kinematic Arrangements:

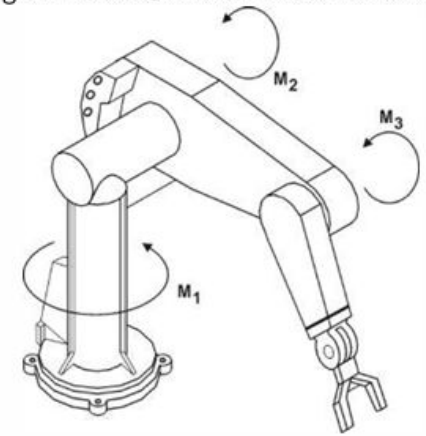
[4] Cylindrical Manipulator (RPP):



[5] Cartesian Manipulator (PPP):

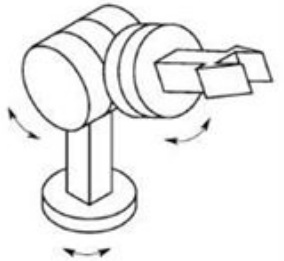


[6] PUMA Manipulator (RRR):
Programmable **U**niversal **M**achine for **A**ssembly:



[7] Spherical Wrist (RRR):

- It is common to attach a spherical wrist to the manipulator end to **allow the orientation of the end-effector**.
- In spherical wrist the axes of the three joints **are intersecting** at the wrist center point.



ROBOT

- Industrial robots are available in a wide variety of sizes, shapes, and physical configurations.
- These configurations are mainly depending on the movement of joints.
- Classification of a robot based on its coordinate system is also referred to as classification by arm configuration or geometric work envelope.
- It tells the volume of reachable coordinates of a point on the end-effecter, rather than its orientations.

ROBOT

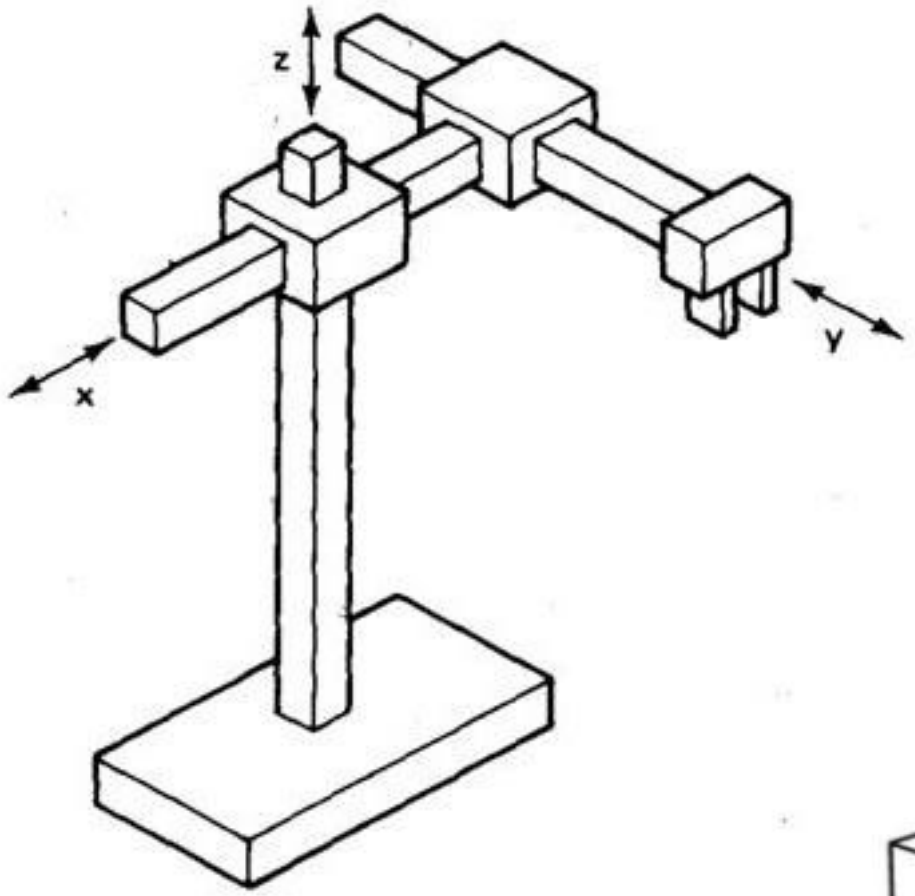
- Those joints are known as ***Prismatic joints*** or Linear joints denoted by P, ***Revolute joints*** denoted by R, and ***Spherical joints*** denoted by S.

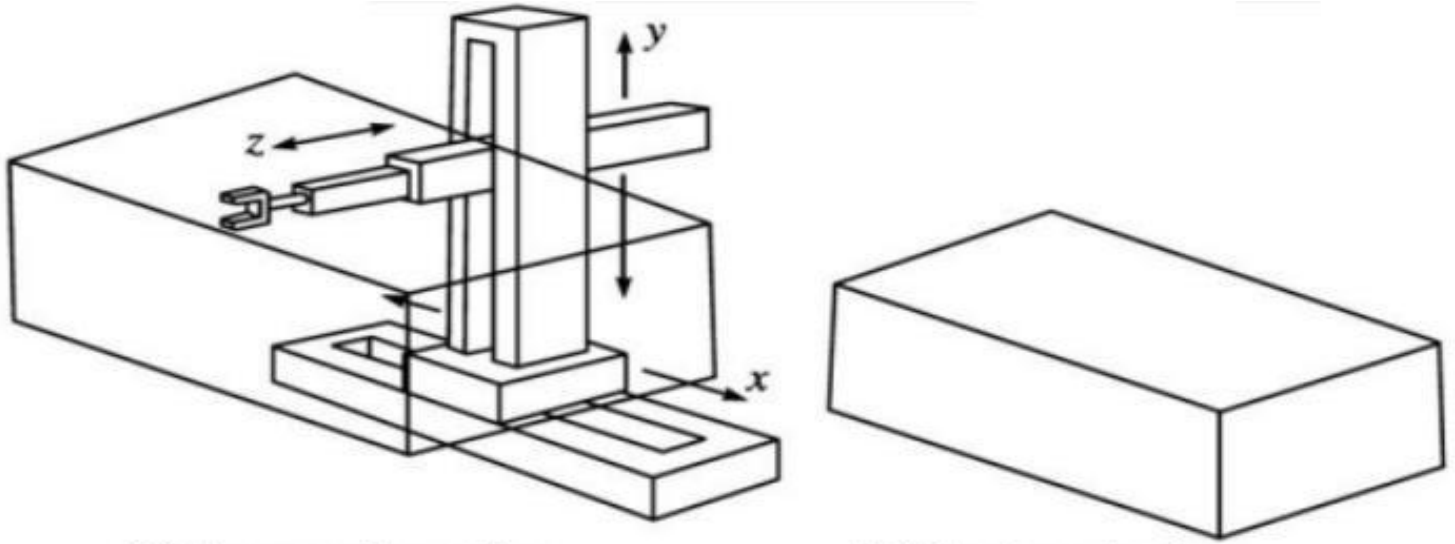
- Four fundamental configuration types are:

1) Cartesian - PPP
2) Cylindrical - RPP
3) Spherical or polar - RRP
4) Articulated or revolute - RRR

Cartesian configuration (PPP)

- When the arm of a robot moves in a rectilinear mode, that is, to the directions of x, y, and z coordinates of the rectangular right-handed Cartesian coordinate system, it is called ***Cartesian or rectangular-coordinate type.***
- The associated robot is then called Cartesian robot.
- It has three prismatic joints (PPP).
- 3P represents ***three Prismatic or Linear joints*** used in the robots.
- By moving the three slides relative to one another, the robot is capable of operating within a ***rectangular work envelope***.



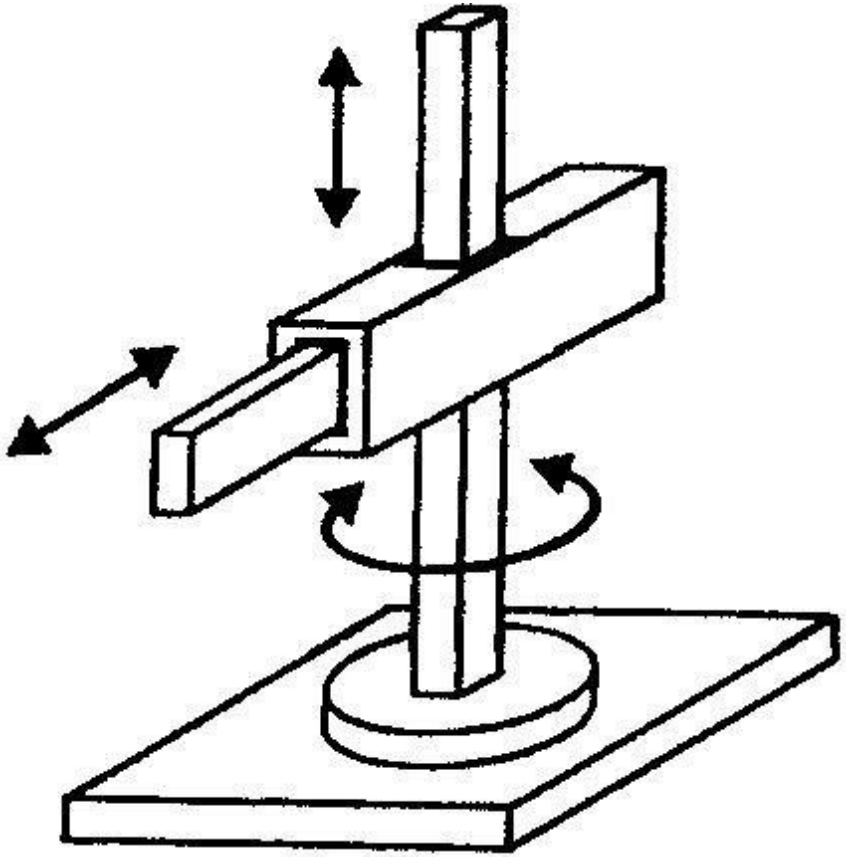


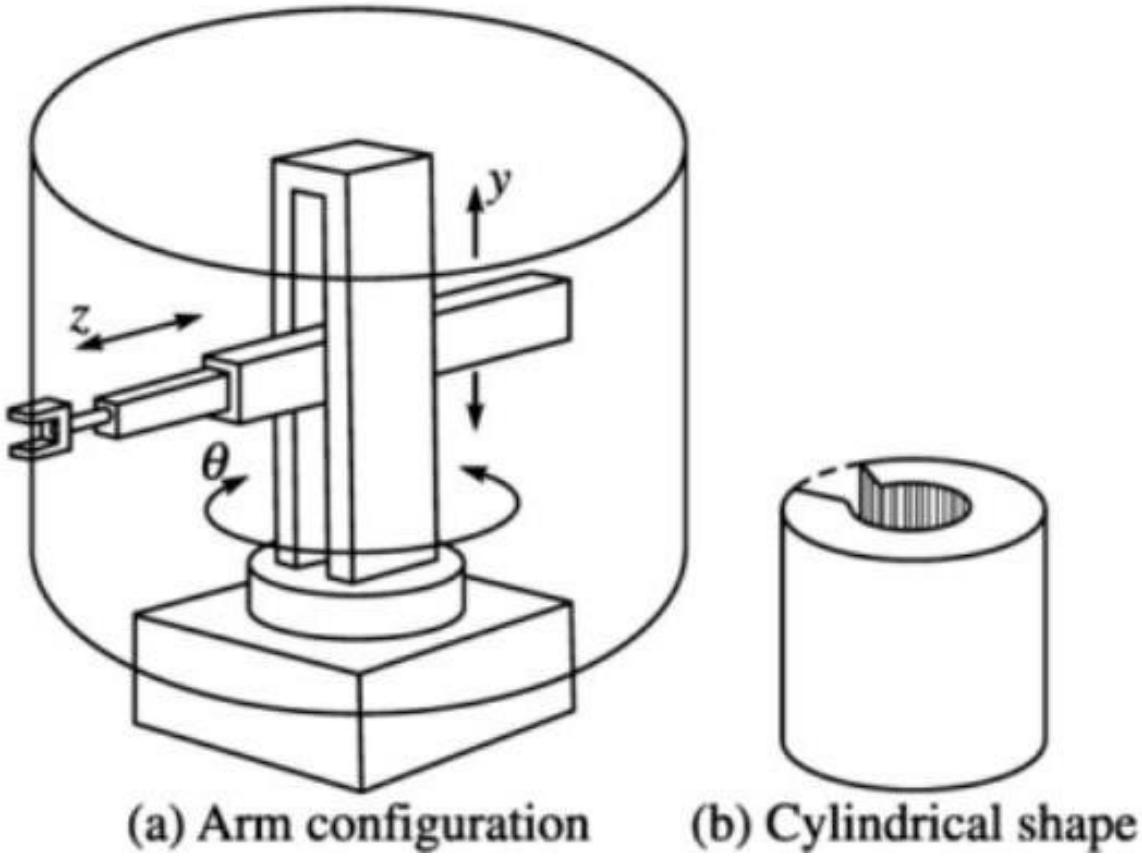
(a) Arm configuration

(b) Rectangular box

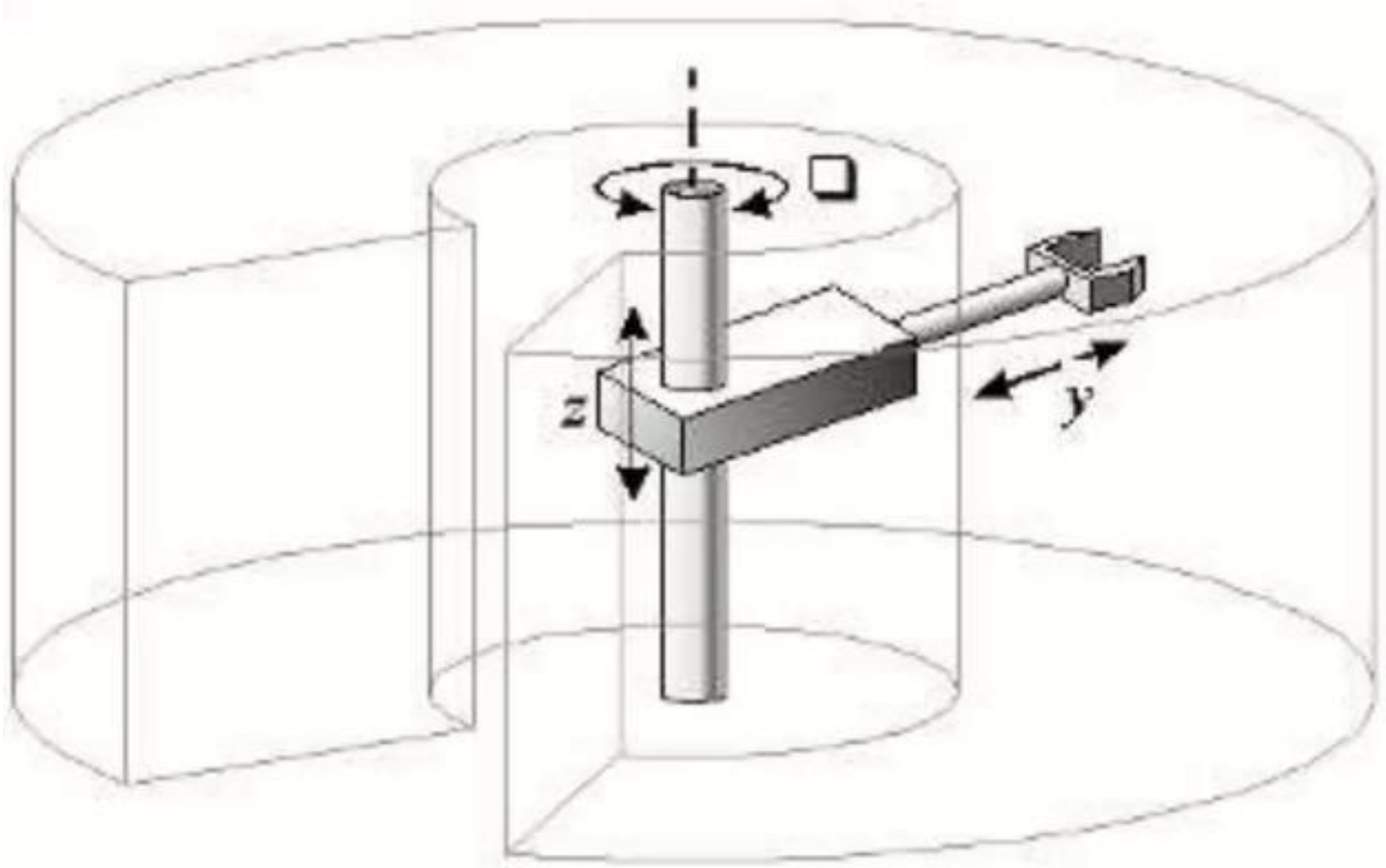
Cylindrical configuration (RPP)

- When the arm of a robot possesses ***one revolute and two prismatic joints*** (RPP), then it is said to have cylindrical configuration.
- Here the arm moves, θ, y, and z, i.e., it has ***base rotation, elevation, and reach***, respectively.
- The first prismatic joint of the Cartesian type is replaced by a revolute one with its axis rotated by 90o about the reach z-axis.
- The points that it can reach conveniently be specified by the cylindrical coordinates, i.e., angle θ, height y, and radius z.
- By rotating the column, the robot is capable of achieving a ***workspace*** that approximates a ***cylinder.***



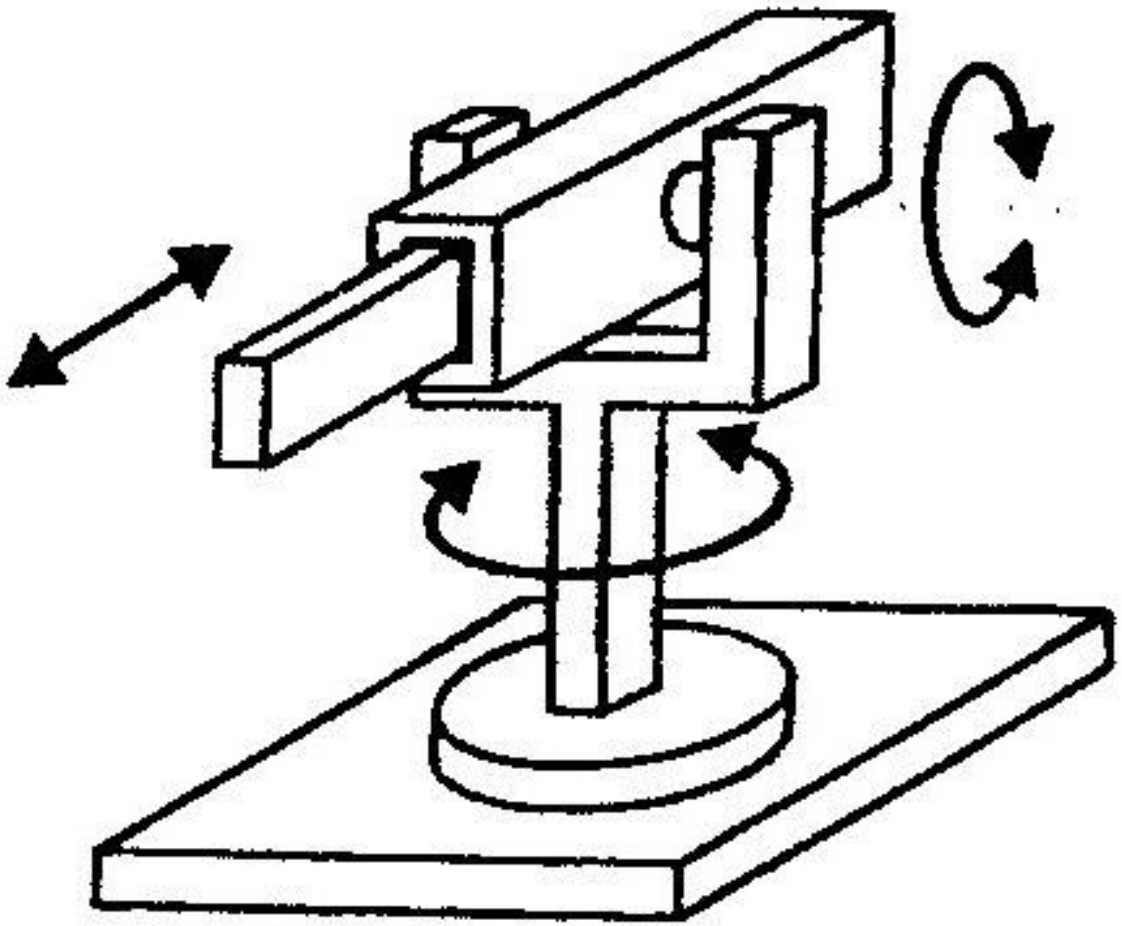


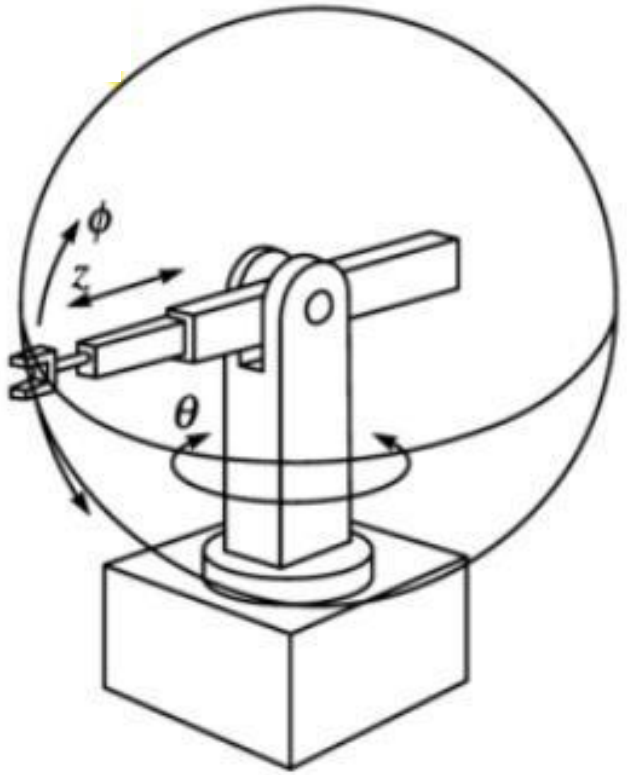
(a) Arm configuration (b) Cylindrical shape


z
y

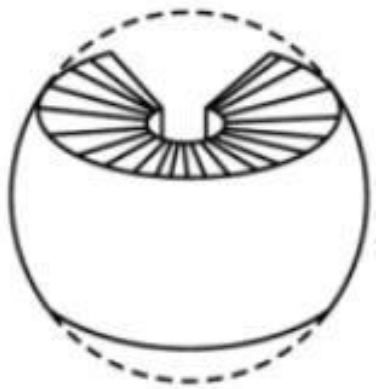
Polar or Spherical configuration (RRP)

- When the arm of a robot can change its configuration by moving its ***two revolute joints and one prismatic joint*** (RRP), then it is has Polar or Spherical configuration.
- The second prismatic joint along the height y of the cylindrical type is replaced by a revolute joint with its axis rotated by 90o about the reach z-axis.
- The arm position is conveniently described by means of the spherical coordinates, i.e., ***θ, φ, and z***.
- The arm movements represent the ***base rotation, elevation angles, and reach***, respectively.
- These joints provide the robot with the capability to move its arm within a ***spherical space***.

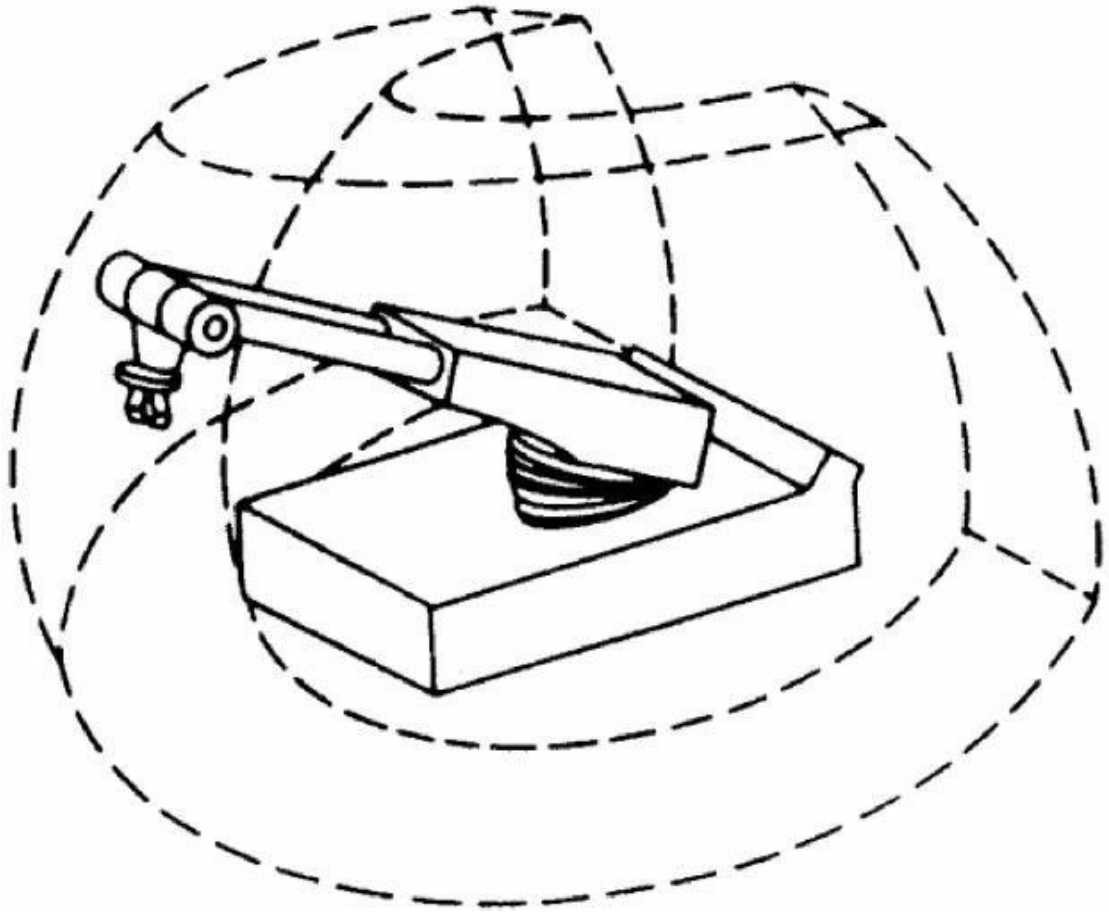




(a) Arm configuration

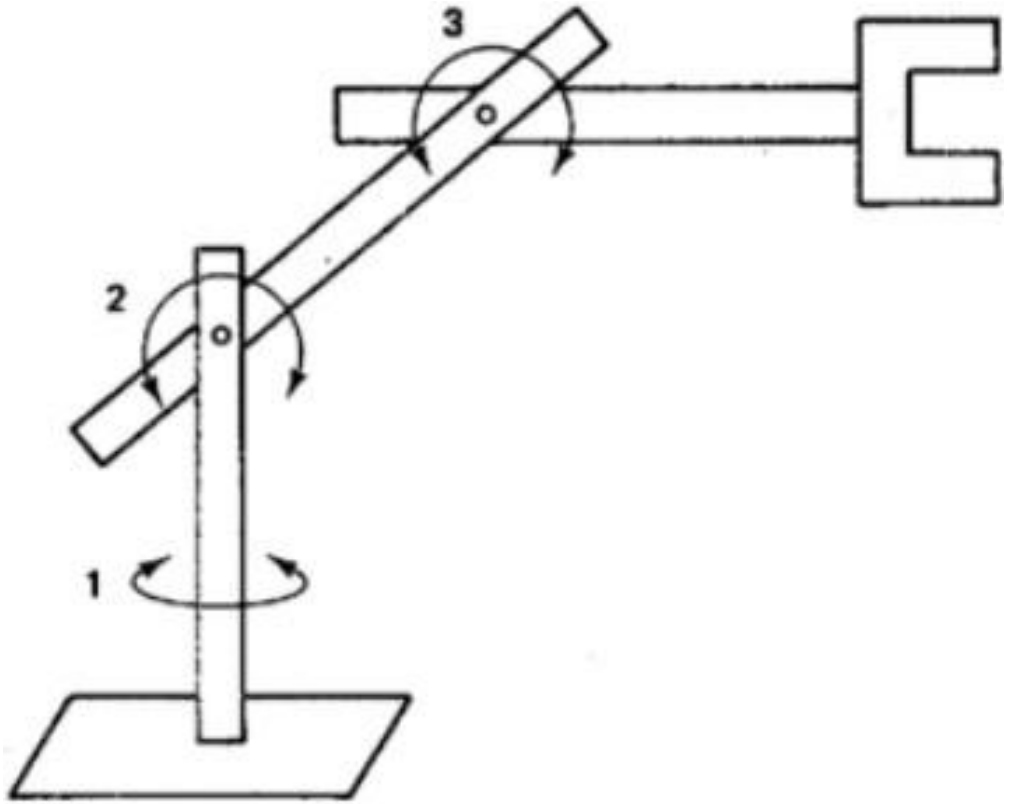


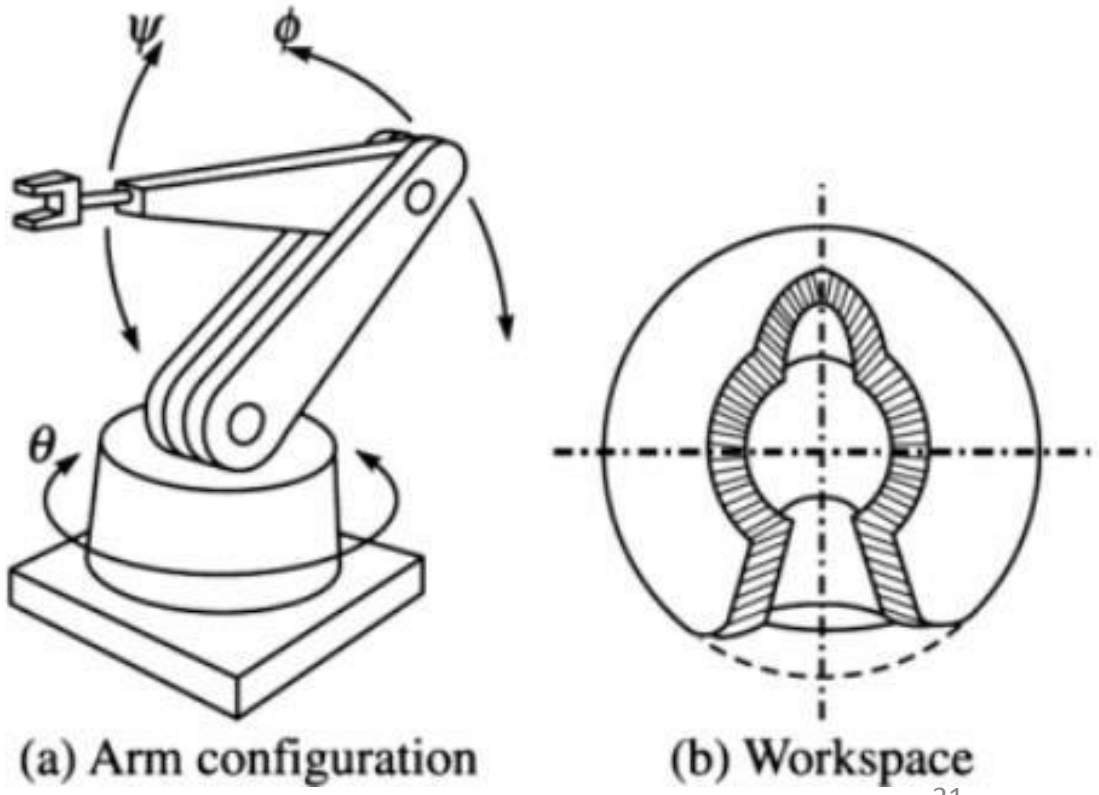
(b) Spherical shape



Articulated or Revolute (RRR) Configuration

- When a robot arm consists of links connected by ***revolute joints only***, then it has articulated configuration.
- The third prismatic joint of the spherical type is also replaced by another revolute joint with its axis rotated by 90° about θ-axis.
- It has ***sphere-like workspace*** and its internal surface is difficult to determine.
- This robot configuration is also sometimes called ***anthropomorphic*** as its anatomy is ***similar to the Human arm***.





(a) Arm configuration

(b) Workspace

Configuration	Advantages	Disadvantages
Cartesian (3 linear axes) x: base travel y: height z: reach	- Easy to visualize - Rigid structure - Easy offline programming - Easy mechanical stops	- Reach only front and back - Requires large floor space - Axes are hard to seal - Expensive
Cylindrical (1 rotation and 2 linear axes) θ : base rotation y: height z : reach	- Can reach all around - Rigid y, z-axes - θ-axis easy to seal	- Cannot reach above itself - Less rigid θ-axis - y, z-axes hard to seal - Won't reach around obstacles - Horizontal motion is circular
Spherical (2 rotating and 1 linear axes) θ: base rotation ϕ: elevation angle z: reach	- Can reach all around - Can reach above or below obstacles - Large work volume	- Cannot reach above itself - Short vertical reach
Articulated (3 rotating axes) θ: base rotation ϕ: elevation angle ψ: reach angle	- Can reach above or below objects - Largest work volume for least floor space	- Difficult to program off-line - Two or more ways to reach a point - Most complex robot

Controller

The controller initiates, terminates, and coordinates the motion of sequences of a robot. also it accepts the necessary inputs to the robot and provides the outputs to interface with the outside world.



The **Controller** is the "brain" of the industrial robotic arm.



It works as a computer and allows the robot to also be connected to other systems.



The program is input with a teach pendant to the controller.



Controller

1. **Point-to-point (PTP) control robot:** is capable of moving from one point to another point. The locations are recorded in the control memory. PTP robots do not control the path to get from one point to the next point. Common applications include component insertion, spot welding, whole drilling, machine loading and unloading, and crude assembly operations.

2. **Continuous-path (CP) control robot:** with CP control, the robot can stop at any specified point along the controlled path. All the points along the path must be stored explicitly in the robot's control memory. Typicalfinishing,applicationgluing,andarc welding operations.

3. **Controlled-path robot:** the control equipment can generate paths of different geometry such as straight lines, circles, and interpolated curves with a high degree of accuracy. All controlled-path robots have a servo capability to correct their path.

Manipulator

- A robotic ***manipulator arm*** is a reprogrammable and multifunctional ***mechanical device*** responsible for moving materials, parts, objects, or tools through programmed motions in order to perform various tasks.

- These mechanical devices are composed ***series of jointed segments*** that form an arm-like manipulator.

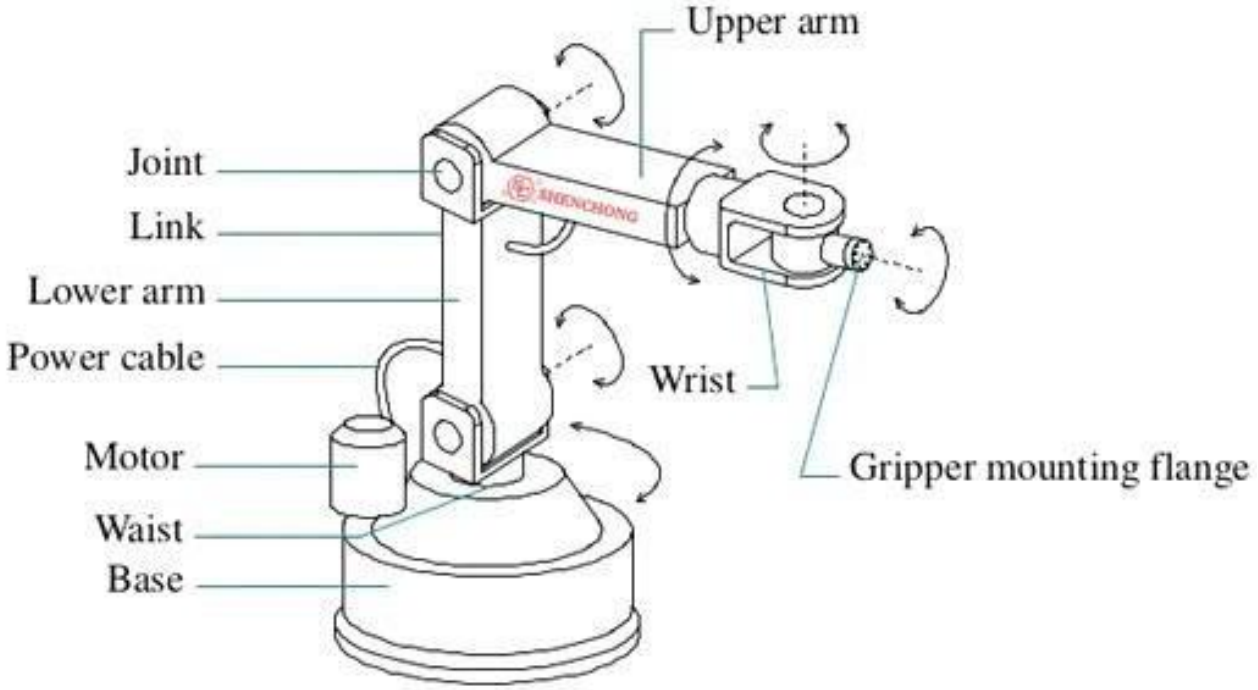
- A robotic manipulator is capable of moving or handling objects automatically depending upon its given number of ***degrees of freedom***. Those degrees of freedom are also known as ***axes***. Each axis of a robotic manipulator correlates to the number of motors within the robot.



Manipulator

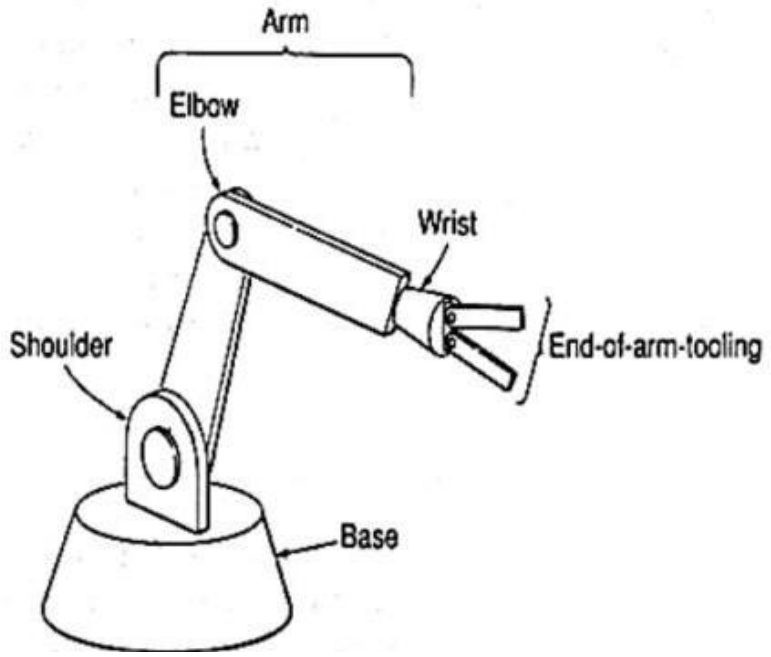
The manipulator, which is the robot's arm, consists of segments jointed together with axes capable of motion in various directions allowing the robot to perform work.





Manipulator

Robotic manipulators are typically divided into two parts:

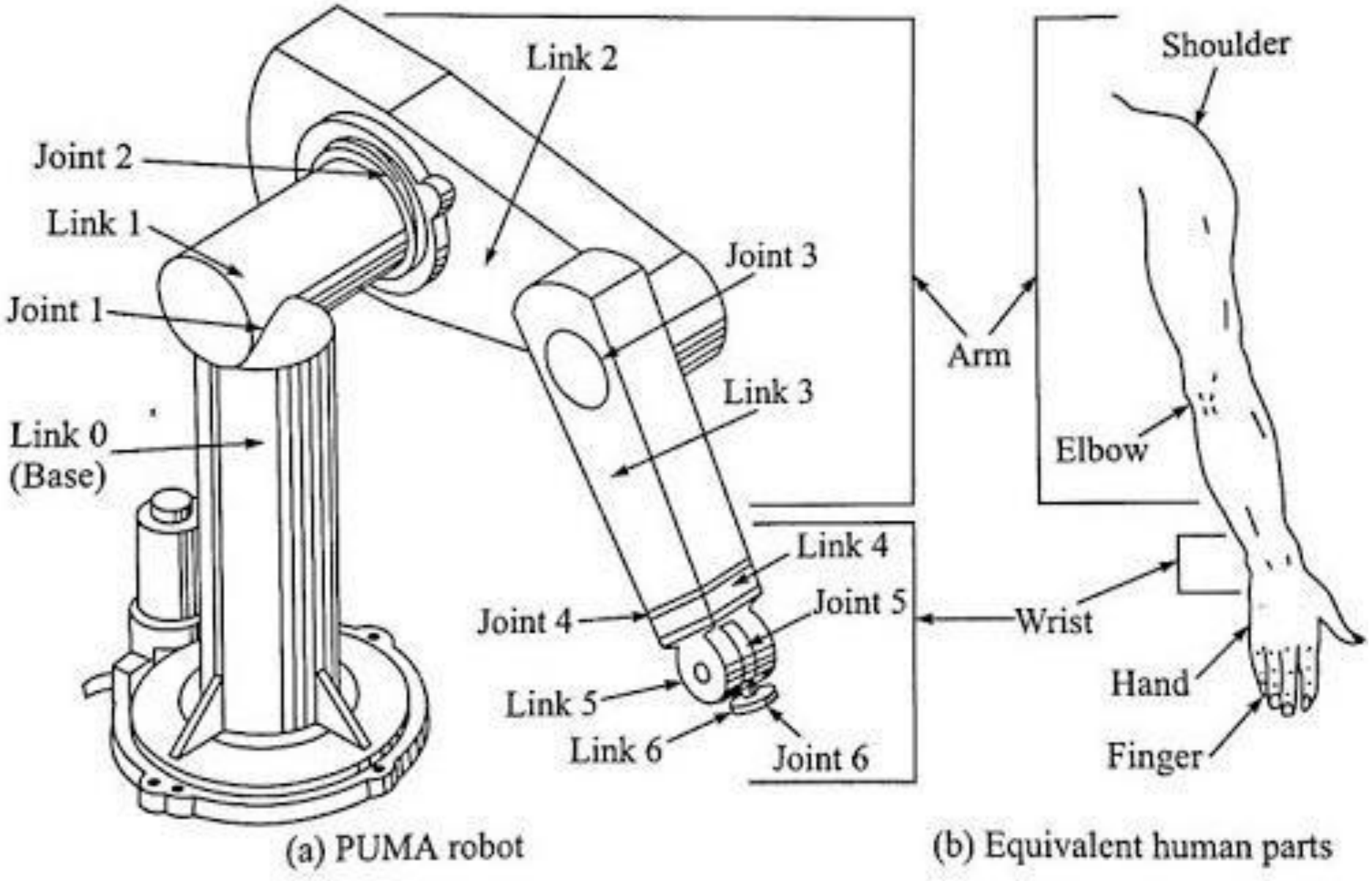


- ***Arm and body*** - The arm and body of the manipulator control the movement of objects within the robot's work envelope.
- ***Wrist*** - While the wrist controls the movement of the end effector, allowing the manipulator to carry out the task it has been programmed.

The body, arm and wrist assembly are called the manipulator.

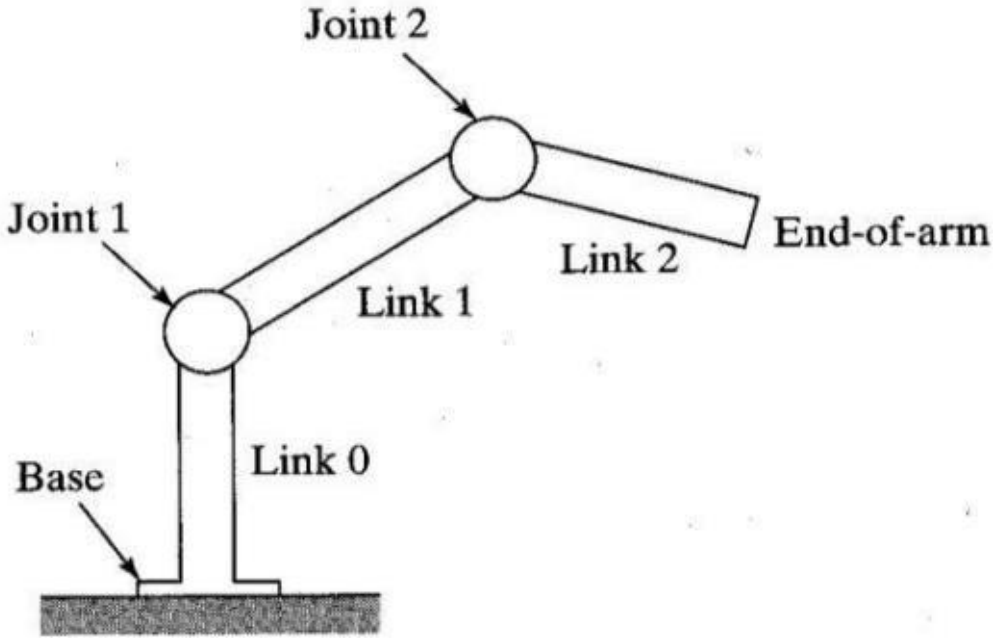
Anatomy of Robotic Manipulator

- Robot Anatomy is concerned with the ***physical construction*** of the body, arm and wrist of the machine.
- Study of different ***joints*** and ***links.***
- Robots are mounted on a ***base*** which is fastened to the floor.
- The ***body*** is attached to the base and the arm assembly is attached to the body.
- At the end of the arm is the ***wrist.***
- Attached to the robot's wrist is a hand or tool called the ***end-effector***. The end-effector is not considered part of the robot's anatomy.
- The arm and body joints are used to position the end-effector (hand or tool) while the wrist joints are used to orient the end-effector


Link 2
Joint 2
Link 1
Joint 3
Joint 1
Link 3
Link 0
(Base)
Link 4
Joint 5
Joint 4
Link 5
Link 6
Joint 6
Arm
Wrist
Shoulder
Elbow
Hand
Finger
(a) PUMA robot
(b) Equivalent human parts

Links

- The individual bodies that make up a robot are called links.
- A link is a ***rigid segment*** of a robot.
- Links are connected by joints.
- By moving (e.g. rotating) the joints, the links can be moved, and the whole robot can be articulated.

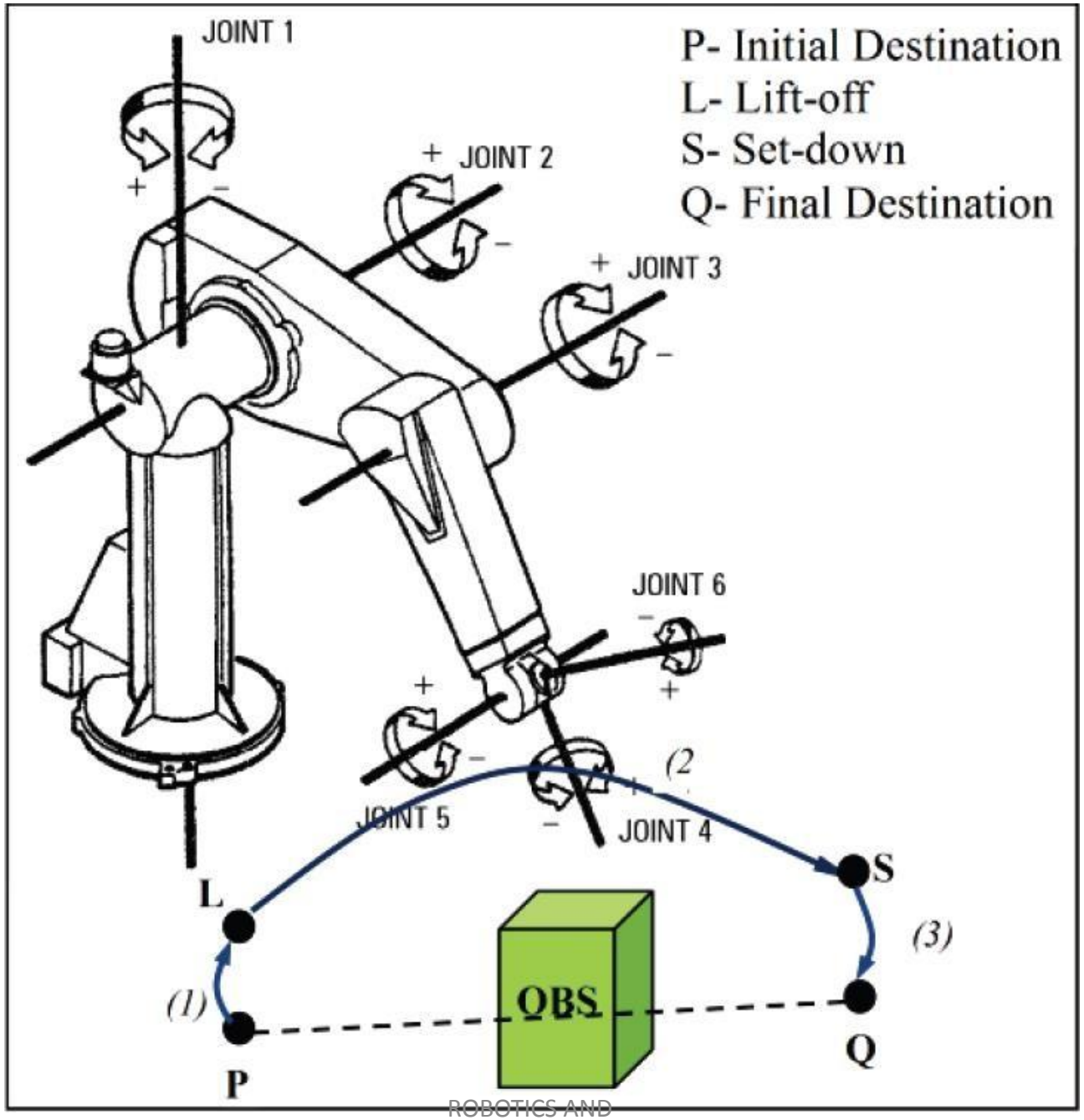


Types of link

I. **Rigid link**: In this type of link, there will not be any deformation while transmitting the motion. For example, the industrial robotic arm is having rigid links, there will not be any deformation while moving the arm.

II. **Flexible link**: In this type of link, there will be a partial deformation while transmitting the motion. One of the examples of flexible links is belt drive.

III. **Fluid link**: In this type of link, motion is transmitted with the help of fluid pressure. Hydraulic actuators, brakes are an example of a fluid link.


JOINT 1
+ JOINT 2
+ JOINT 3
JOINT 6
JOINT 5
JOINT 4
P- Initial Destination
L- Lift-off
S- Set-down
Q- Final Destination
L
S
Q
P
(1)
(2
(3)
OBS

Joints

- A joint ***couples two links*** and provides physical constraints on the relative motion between the links.
- A joint is a ***connection*** between two or more links, which allows some motion, or potential motion, between the connected links.
- Joints can be classified as:

TRANSLATIONAL MOTION

- Linear joint (type L)
- Orthogonal joint (type O)

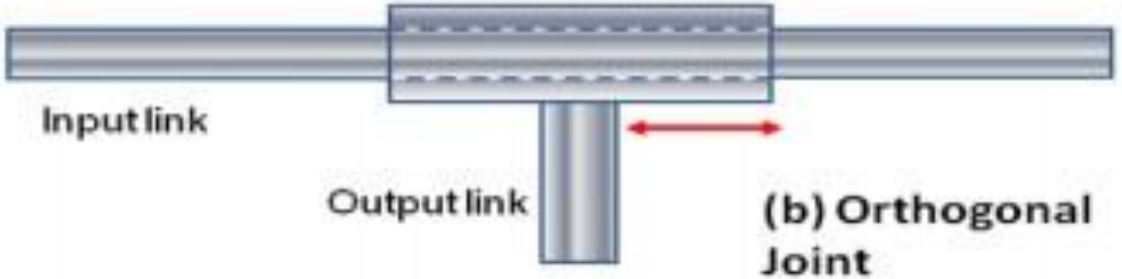
ROTARY MOTION

- Rotational joint (type R)
- Twisting joint (type T)
- Revolving joint (type V)

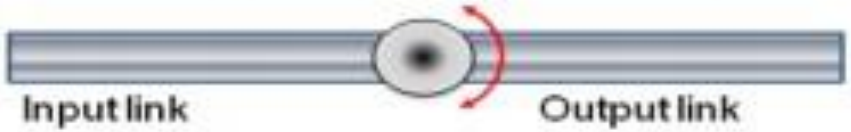
Joints



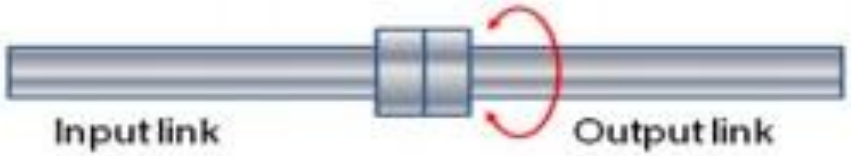
(a) Linear Joint



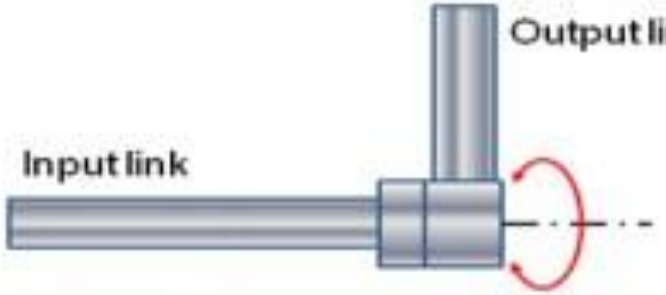
(b) Orthogonal Joint



(c) Rotational Joint



(d) Twisting Joint



(e) Revolving Joint

Joints

a) Linear joint (type L joint)

The relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links being parallel.

b) Orthogonal joint (type U joint)

This is also a translational sliding motion, but the input and output links are perpendicular to each other during the move.

c) Rotational joint (type R joint)

This type provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links.

Joints

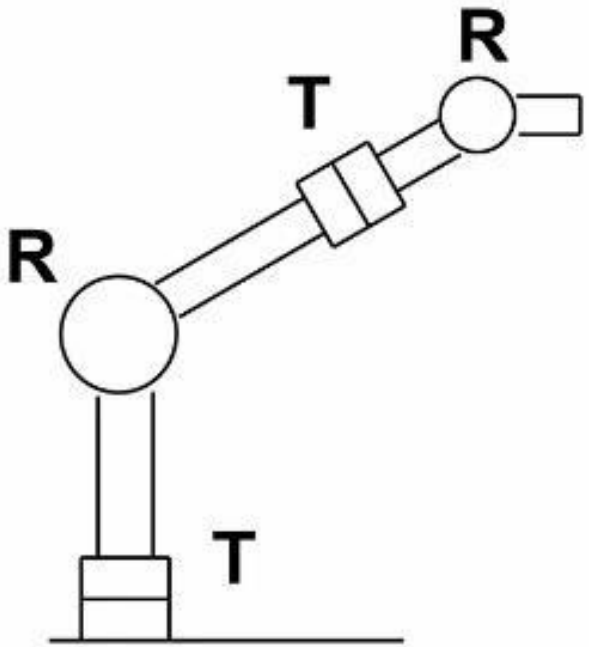
d) Twisting joint (type T joint)

This joint also involves rotary motion, but the axis or rotation is parallel to the axes of the two links.

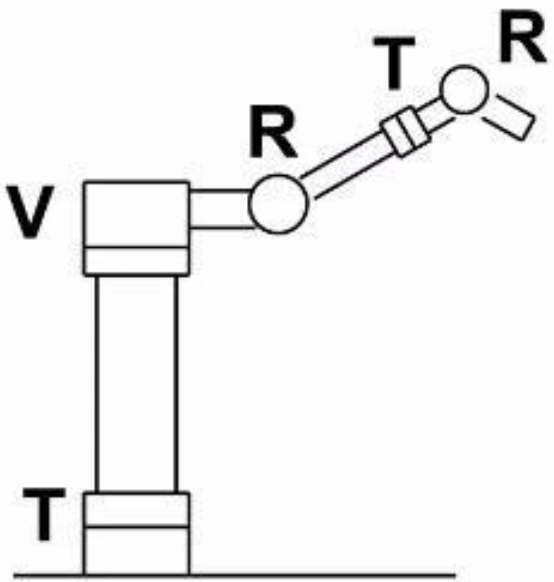
e) Revolving joint (type V-joint, V from the "v" in revolving)

In this type, axis of input link is parallel to the axis of rotation of the joint. However the axis of the output link is perpendicular to the axis of rotation.

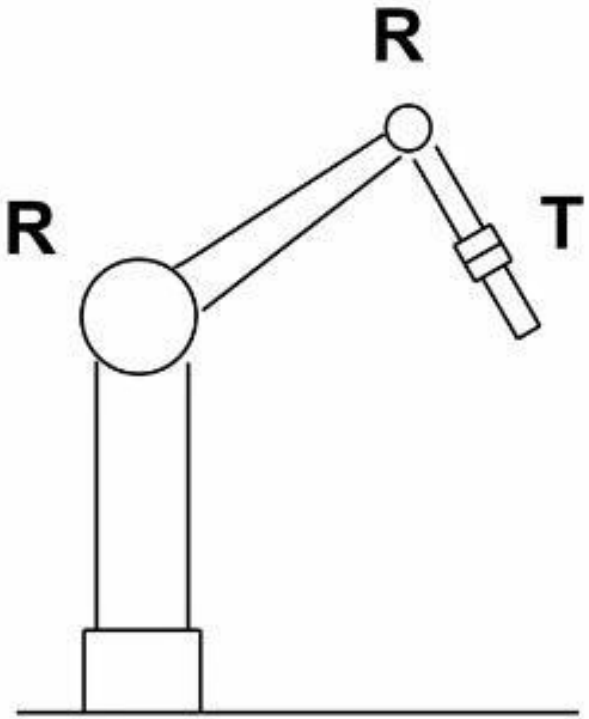
Joints



(a) TRT:R



(b) TVR:TR



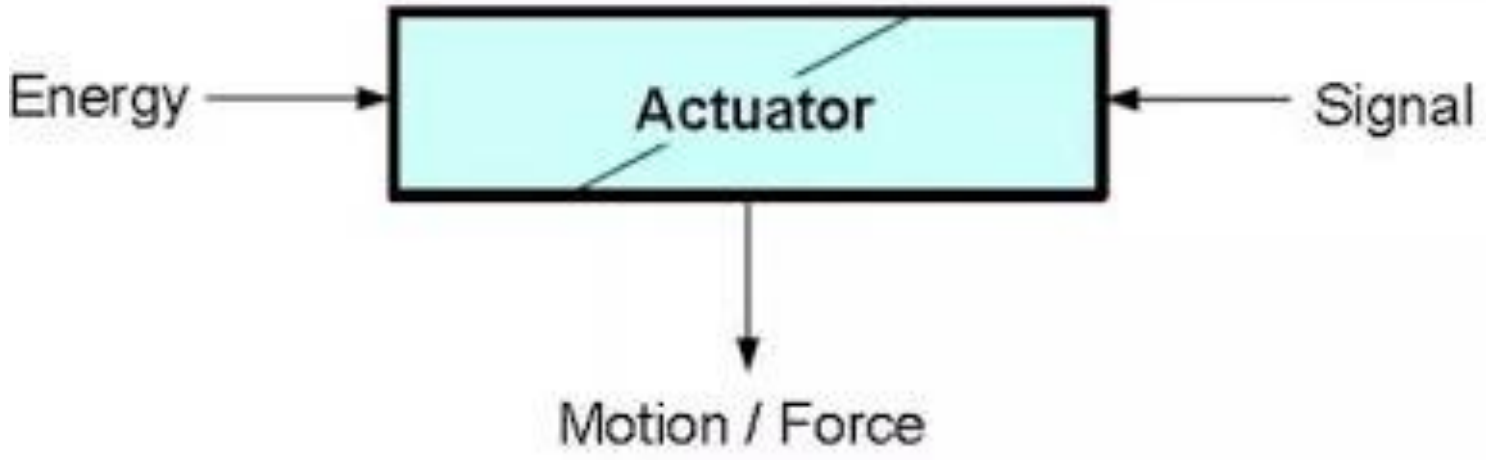
(c) RR:T

Actuators

Actuators are the devices which provide the actual motive force for the robot joints.

The actuators of a robot actually ***provide motion*** to the manipulator links and end-effector.

They are the '***Muscles of the Robot'.***



Actuators

- They are classified as pneumatic, hydraulic or electric based on their principle of operation.

- Pneumatic robots (use compressed air as driving force) are used in light assembly or packing work but are not suitable for heavy-duty tasks or where speed control is necessary.

- Hydraulic robots are used in heavy payload applications because of their high power to size ratios.

Actuators

1. ***Electric Drives -*** The primary input power supply is the ***electric energy*** from the electric distribution system.

2. ***Hydraulic Drives -*** They transform ***hydraulic energy*** stored in a reservoir into mechanical energy by means of suitable pumps.

3. ***Pneumatic Drives -*** They utilize ***pneumatic energy***, i.e., ***compressed air***, provided by a compressor and transform it into mechanical energy by means of pistons or turbines.

Types of

Robotic actuators are classified into two types according to the requirements of motion

For Linear Motion:

a) Linear Actuators
b) Solenoid Actuators

For Rotational Motion:

a) DC Motor Actuators
b) Servo Actuators
c) Stepper Motor Actuators





Types of

Linear Actuators

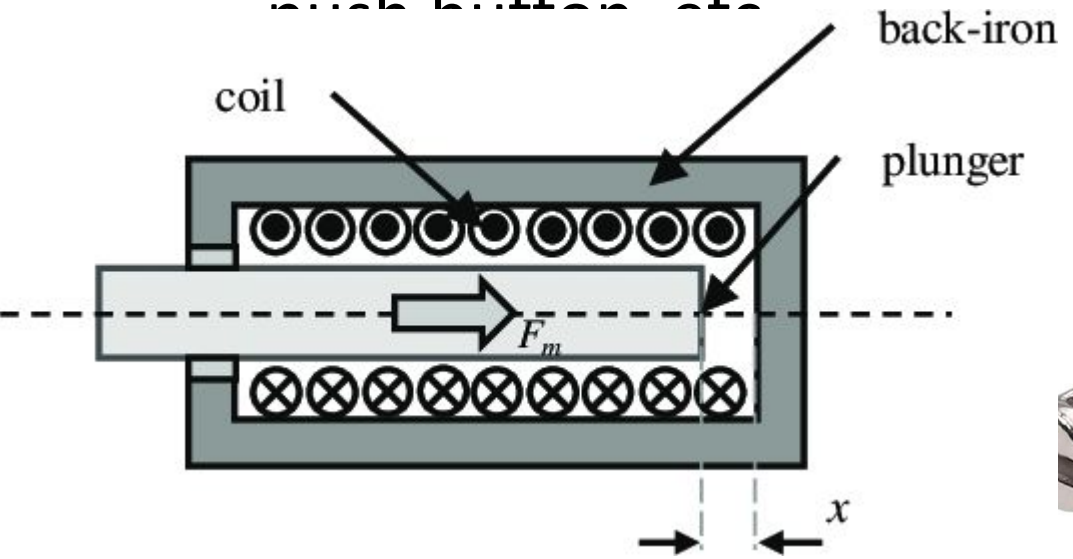
- Linear actuators in robotics are used to ***push or pull*** the robot like move forward or backward & arm extension.
- This actuator's active end is simply connected to the robot's lever arm to activate the such motion.
- These actuators are used in a number of applications in the robotics industry.



Types of

Solenoid Actuators

- Solenoid actuators are special-purpose ***linear actuators*** that include a solenoid latch that works on ***electromagnetic activity***.
- These actuators are mainly used for controlling the motion of the robot and also perform different activities such as a start & reverse, latch, push button, etc.





Types of

- **DC Motor Actuators**
  - DC motor actuators are generally used for turning robotic motion.
  - These actuators are available in different sizes with torque generation capability.
  - It can be utilized for changing speed throughout rotating motions.
  - By using these actuators, different activities like robotic drilling & robotic drive train motion are performed.



Types of

Servo Actuators

- Servo motor actuators in robotics are mainly used to control & monitor rotating motion.
- These are very superior DC motors that allow 360 degrees of rotation, but, continuous revolution is not compulsory.



Types of

Stepper Motor Actuators

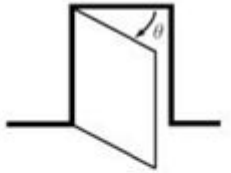
- Stepper motor actuators are helpful in contributing to ***repetitive rotating*** activities within robots.
- These types of actuators are a combination of both DC & servo motor actuators.
- These stepper motor actuators are utilized in automation robots where repeatability of activity is necessary.

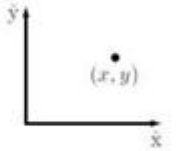


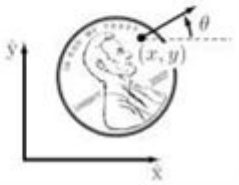
Degrees of Freedom (DoF)

Configuration Space:

- ✔ **Robot's configuration: a specification of the positions of all points of the robot.**
- ✔ **Since the robot is rigid, only a few numbers are needed to represent its configuration.**
- ✔ **The n-dimensional space containing all possible configurations of a robot is called the configuration space (C-space)**

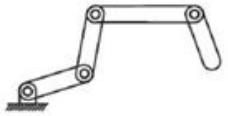






Robot's Degrees of Freedom (n):

- ✔ **Is the smallest number n of real-valued coordinates needed to represent the robot's configuration.**

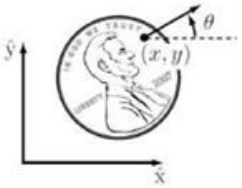


Rigid Body DoF (m):

- ✔ **A rigid body in three-dimensional space, which we call a spatial rigid body, has six degrees of freedom, m = 6 (three for position and three for orientation).**



- ✔ **A rigid body moving in a two-dimensional plane, which we call a planar rigid body, has three degrees of freedom, m = 3 (two for position and one for orientation)**

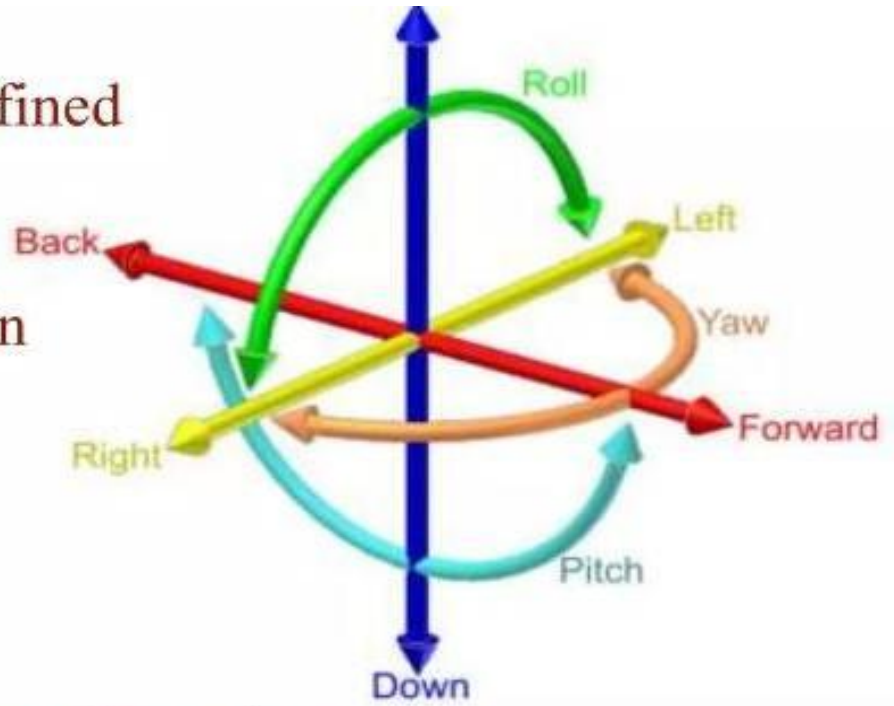


Degree of Freedom

- Basically refers to the number of ways in which an object can move.
- ***'Motion capability'***
- The degrees of freedom of a robot typically refer to the ***number of movable joints*** of a robot.
- For each Degree of Freedom a joint is required.
- A robot with three movable joints will have three axis and three degrees of freedom, a four axis robot will have four movable joints and four axis, and so on.

Degree of Freedom

- The ***degrees of freedom*** (dof) of a rigid body is defined as the number of independent movements it has.
- Higher number dof indicates an increased flexibility in positioning a tool.
- For each degree of freedom a joint is required.



- The degrees of freedom located in the arm define the configuration.
- Three degrees of freedom located in the wrist give the end effector all the flexibility.

Degree of Freedom

Translation – The motion along a path.

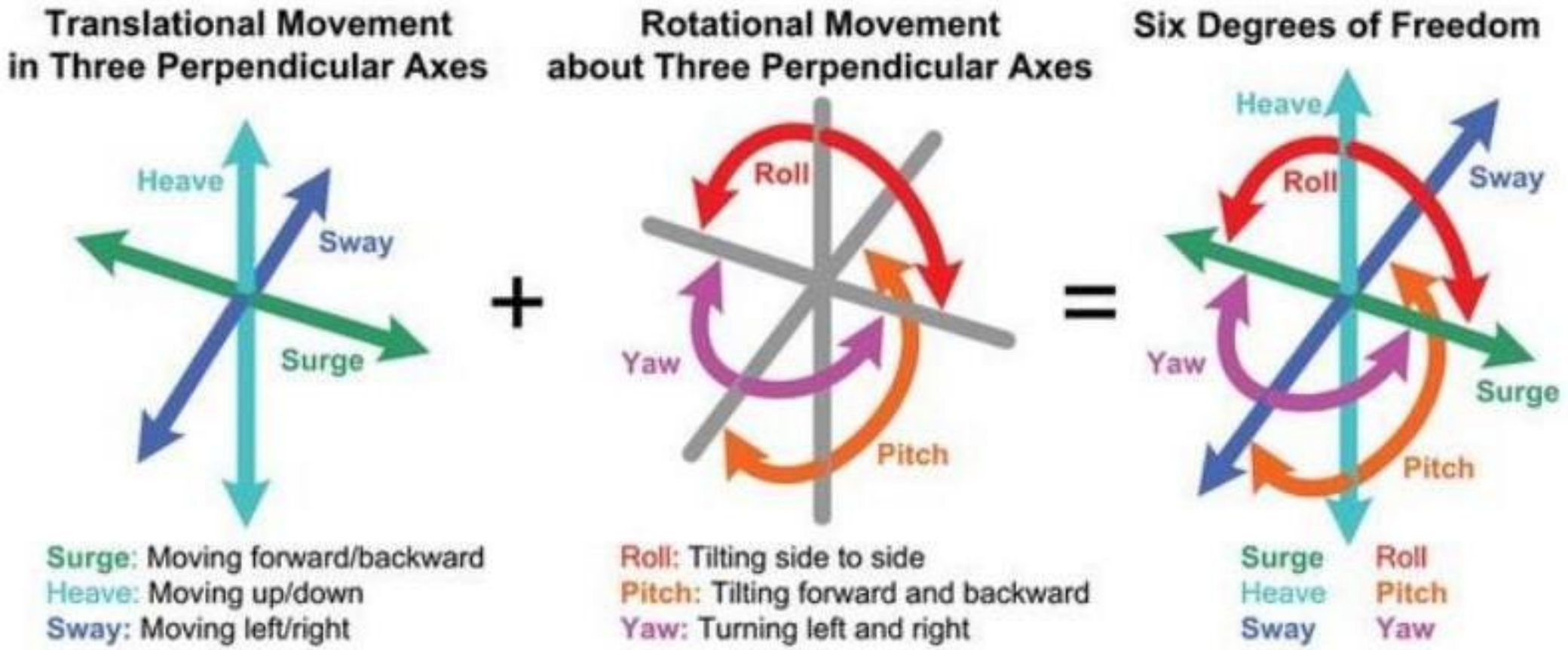
- Forward - Backward
- Left – Right
- Up – Down

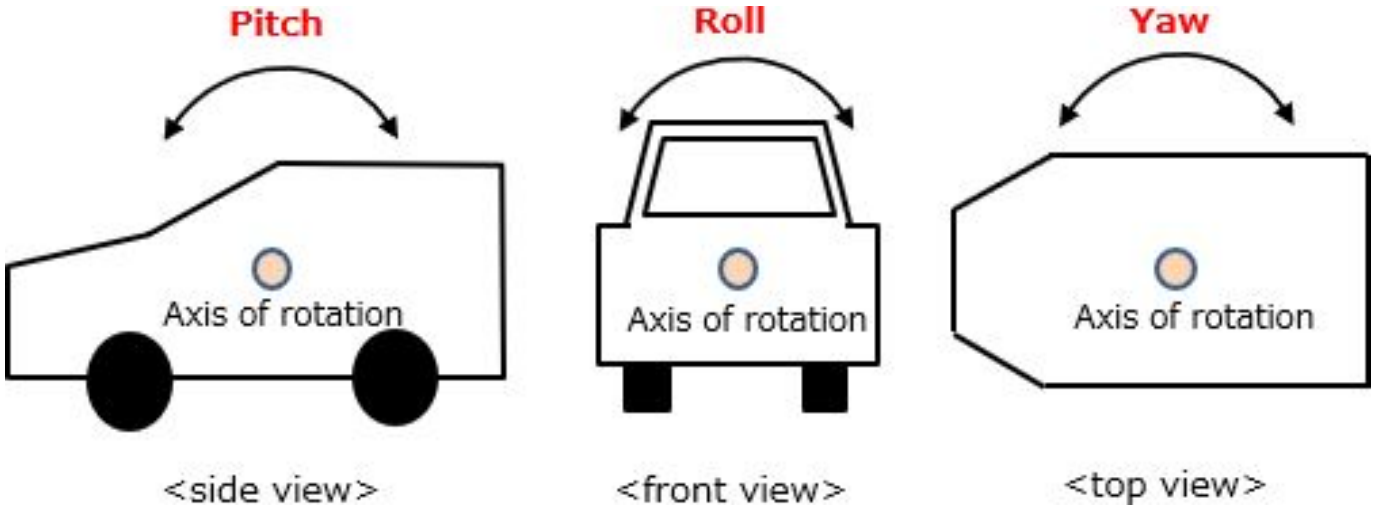
Rotation - The motion of an object around a circular path, in a fixed orbit.

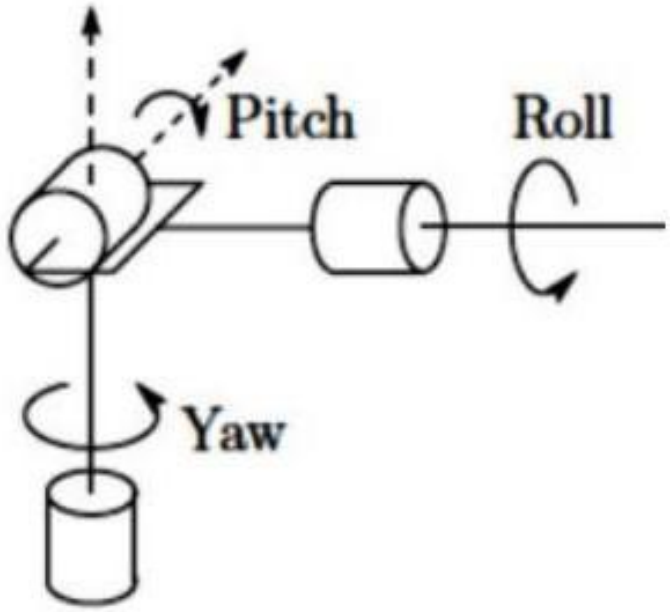
- Pitch
- Yaw
- Roll

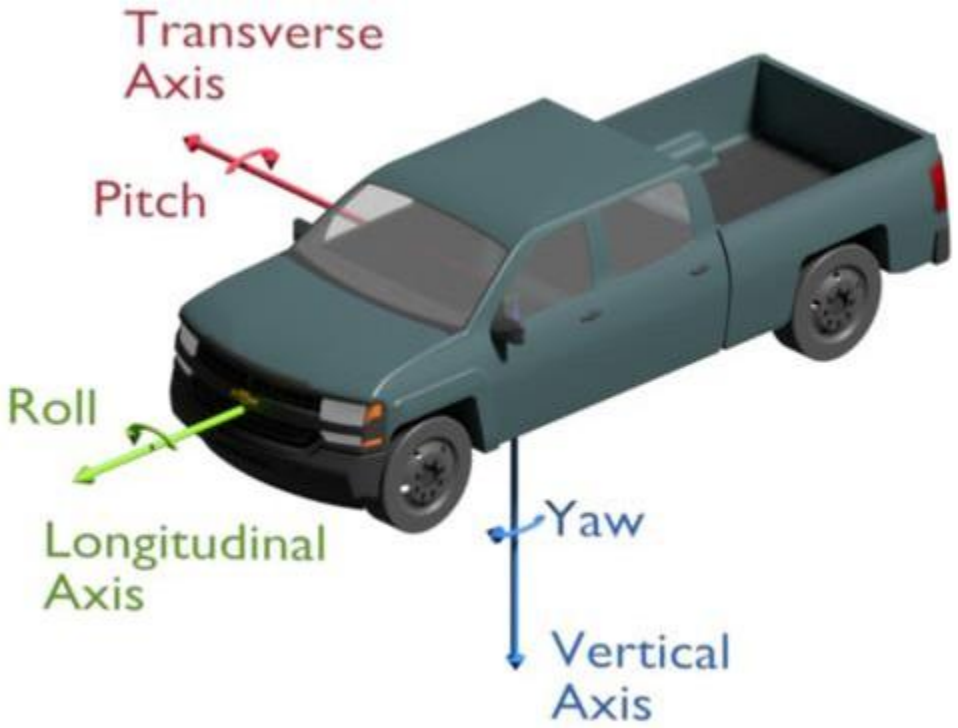
These are the 6 degrees of freedom.

Degree of Freedom




Pitch
Roll
Yaw
Axis of rotation
Axis of rotation
Axis of rotation
<side view>
<front view>
<top view>

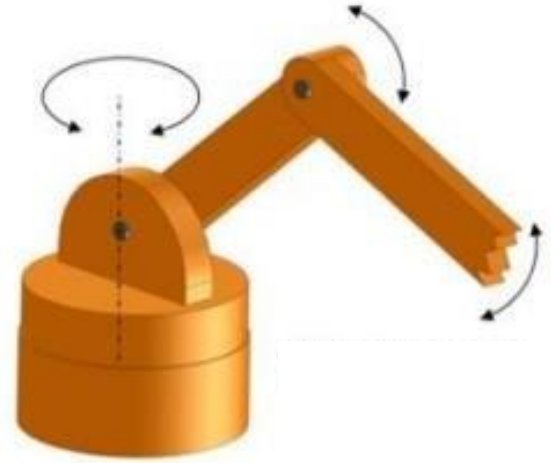

Pitch
Roll
Yaw

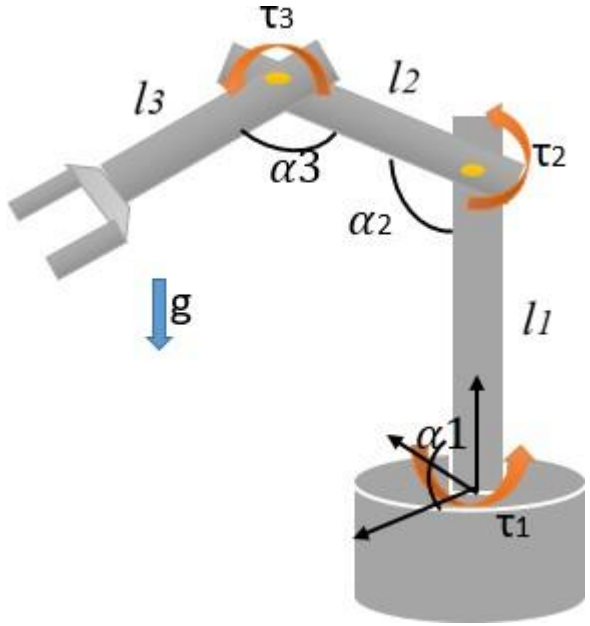

Transverse Axis
Pitch
Roll
Longitudinal Axis
Yaw
Vertical Axis

Different degree of freedom in robotic arm

3 Degree of freedom

- This is a robot that has ***three joints*** that work alongside each other to effect movement in a robotic arm.
- Used for all kinds of robotic work from welding, pick and place, machine tending handling etc.

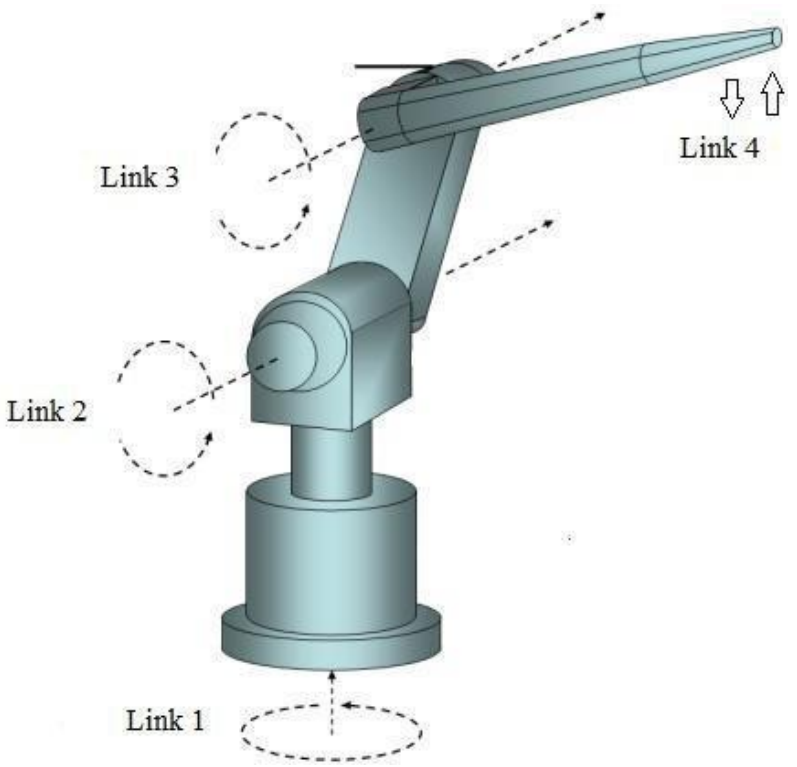




Different degree of freedom in robotic arm

4 Degree of freedom

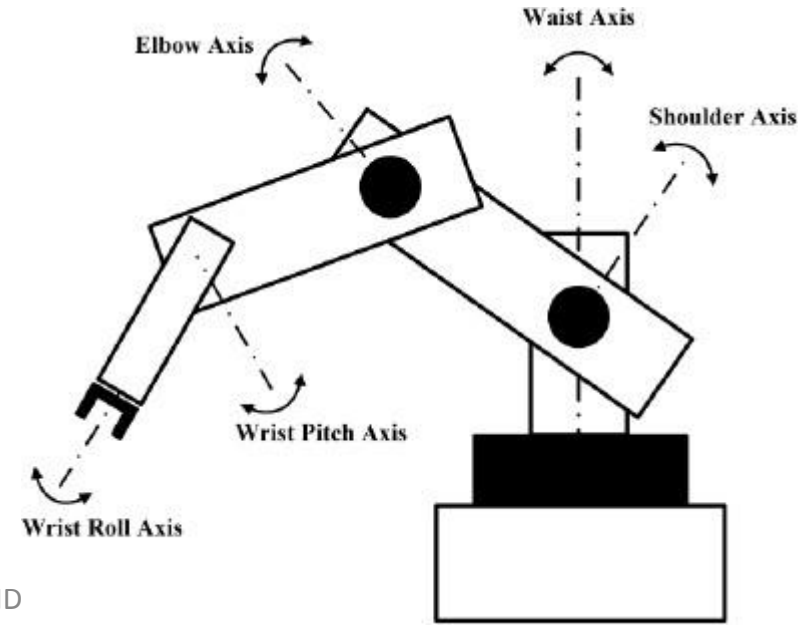
- This is a robot that has ***four axes or joints***.
- The last axis is located near the base of the robot, and it provides the movement and the stability needed for the entire robotic arm to function correctly.
- This type of robot is used in palletizing, machine loading, pick and place, automated packaging, among many other roles.

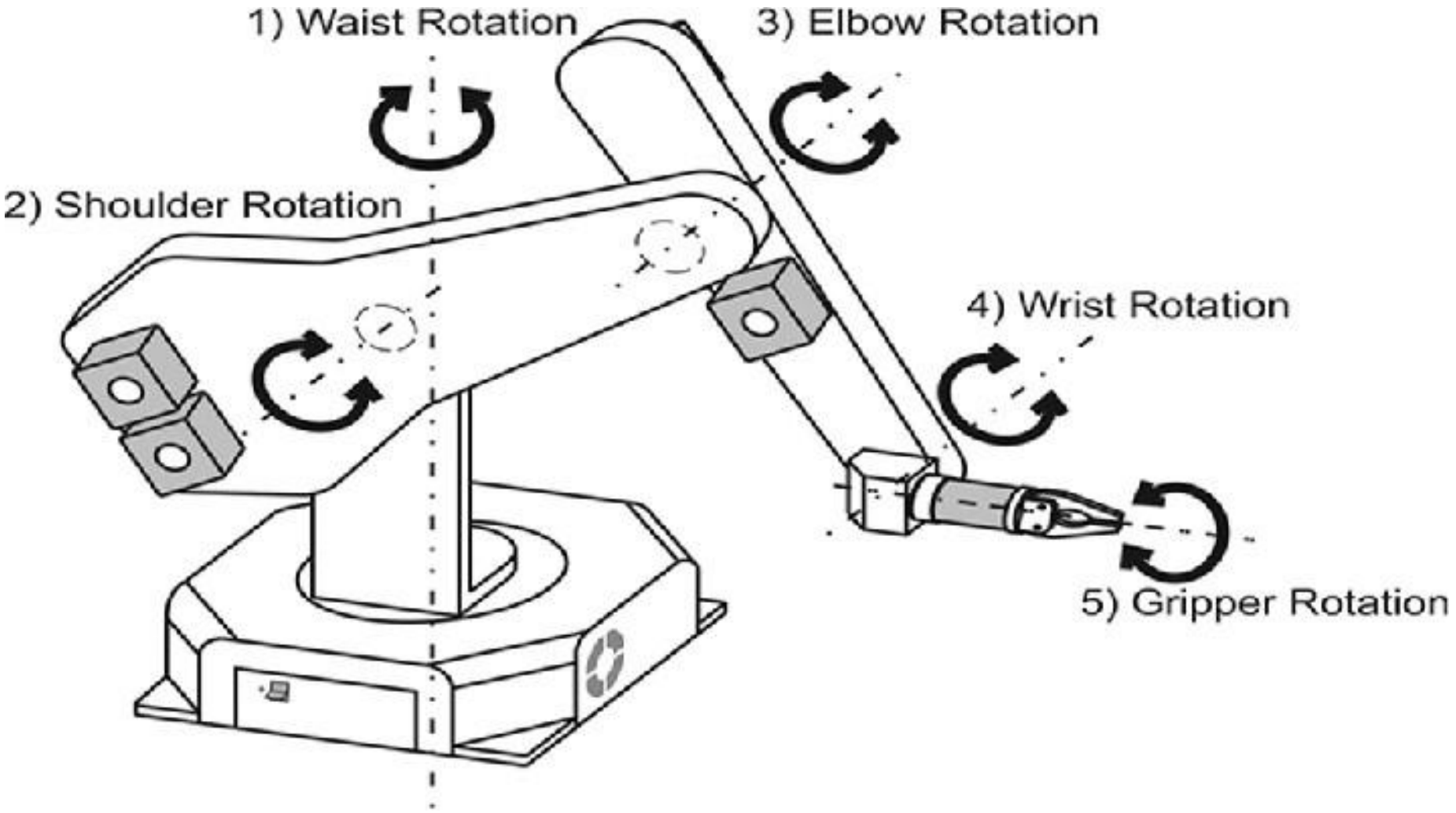


Different degree of freedom in robotic arm

5 Degree of freedom

- This is a robotic arm that has ***five joints***, including a manipulator, a servo-motor actuator, and corresponding arm components like the arm, the elbow, and the wrist.
- It is more complex than the previous two and can handle more due to the presence of more joints.

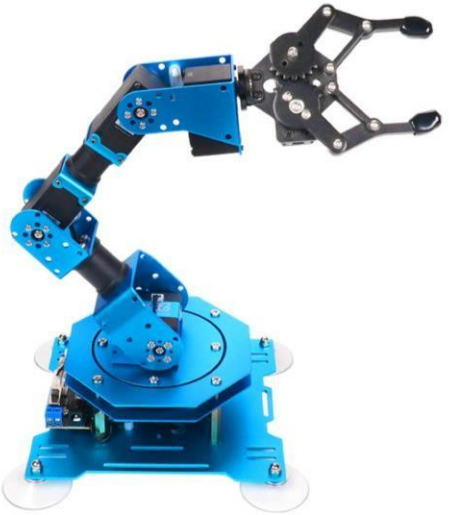



1) Waist Rotation
3) Elbow Rotation
2) Shoulder Rotation
4) Wrist Rotation
5) Gripper Rotation

Different degree of freedom in robotic arm

6 Degree of freedom

- A 6 degrees of freedom robotic arm is a robotic arm that has the ability to move in ***six different directions***.
- This include ***translation*** along the x, y, and z axes, as well as ***rotation*** around these same axes.
- The 6 DOF robotic arm is ***stronger and faster*** and can handle even bigger roles in manufacturing.







Degrees of Freedom (DoF) : Gr¨ubler's formula:

The number of degrees of freedom of a mechanism with links and joints can be calculated using Grubler's formula:

DoF = (sum of freedoms of the bodies)- (number of independent constraints)

If a mechanism has N links including ground, and J joints, its DoF is determined by:

$$\text{DoF} = m(N-1-J) + \sum_{i=1}^{J} f_i$$

Note:

m=3 for planar and m = 6 for rigid mechanisms.

f_i is the number of freedoms provided by joint i.

End-effector

- The end-effector is the device that allows the robotic manipulator to perform tasks and interact with workpieces. Without an end-effector a robot is not very useful.
- The industrial robotic arm may be programmed to follow an application path, but if it does not have a device to interact with workpieces, then it cannot automate the application.
- Robot end-effectors are integrated with the end of the robot wrist.
- They may be powered hydraulically, electrically, or pneumatically. Other terms used for end-effectors include EOAT (end of arm tooling) and manipulator.

End-Effector Types

End-effectors can be classified into three categories. These include:

- Grippers, Process tooling, Sensors.

Type	Examples	Uses & applications
Grippers	Mechanical (jaws/fingers) Vacuum (suction cups) MagneticSoft (silicone)	Pick-and-place, assembly, packaging, food handling, fragile or irregular objects
Process tools	Welding torches Painting nozzles Deburring and finishing tools	Automotive welding, painting, edge finishing, cutting, dispensing adhesives or sealants
Sensors & vision	Cameras Force/torque sensors LiDAR Tactile sensors	Quality inspection, barcode scanning, assembly guidance, navigation, collaborative robots
Specialized tools	Surgical instrumentsCustom palletizing grippers3D printing extrudersSorting tools	Healthcare surgery, logistics, additive manufacturing, agriculture, recycling or sorting

Grippers

- Grippers are end-effectors, which are used to grasp an object or a tool, e.g., a grinder, and hold it.
- Tasks required by the grippers are to hold workpieces and load/unload from/to a machine or conveyer.
- Grippers can be mechanical in nature using a combination of mechanisms driven by electric, hydraulic, or pneumatic powers, as explained in earlier sections.
- Grippers can be classifi ed based on the principle of grasping mechanism
- Classified as:
    - Mechanical Grippers
    - Magnetic Grippers
    - Vacuum Grippers
    - Adhesive Grippers
    - Hooks, Scoops, and Others

Mechanical Grippers

- Mechanical grippers have their jaw movements through pivoting or translational motion using a transmission element, e.g., linkages or gears, etc.
- The function of a gripper mechanism is to translate some form of power input, be it electric, hydraulic or pneumatic, into the grasping action of the fi ngers against the part.
- The motion of the fi ngers in mechanical grippers is obtained by means of any of the following transmission elements: (i) Linkage; (ii) Gear and rack;(iii) Cam; (iv) Screw; and (v) Cable and pulley, etc.

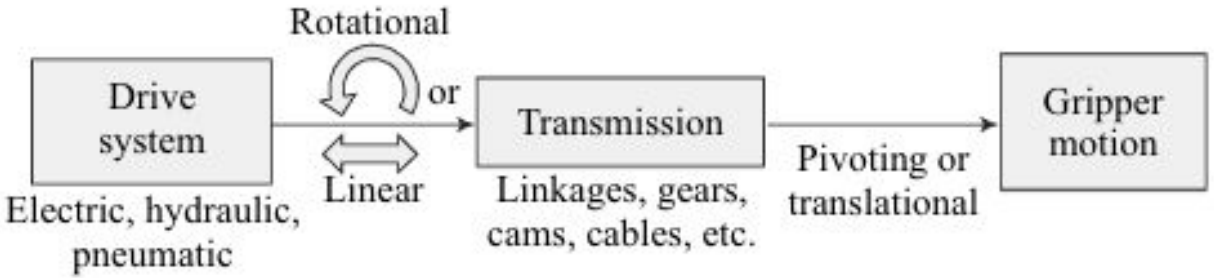
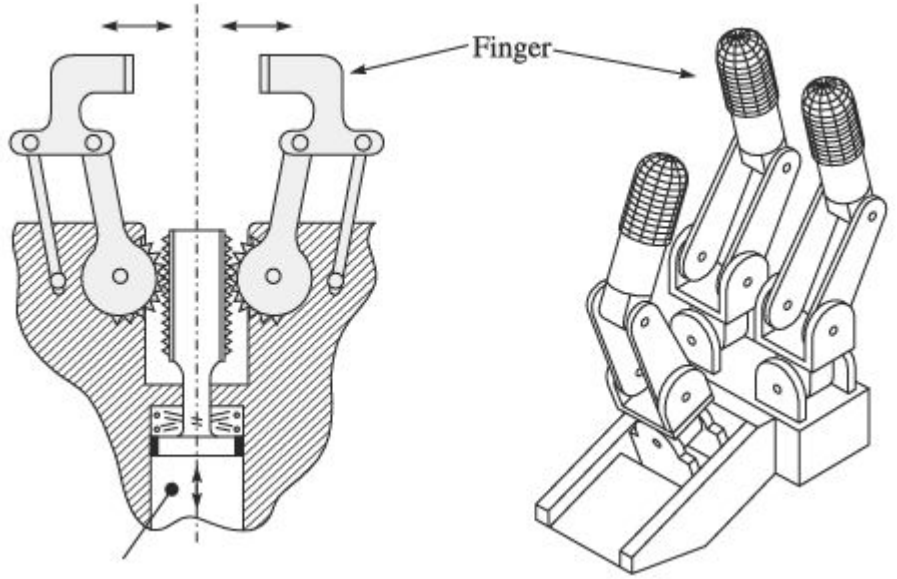
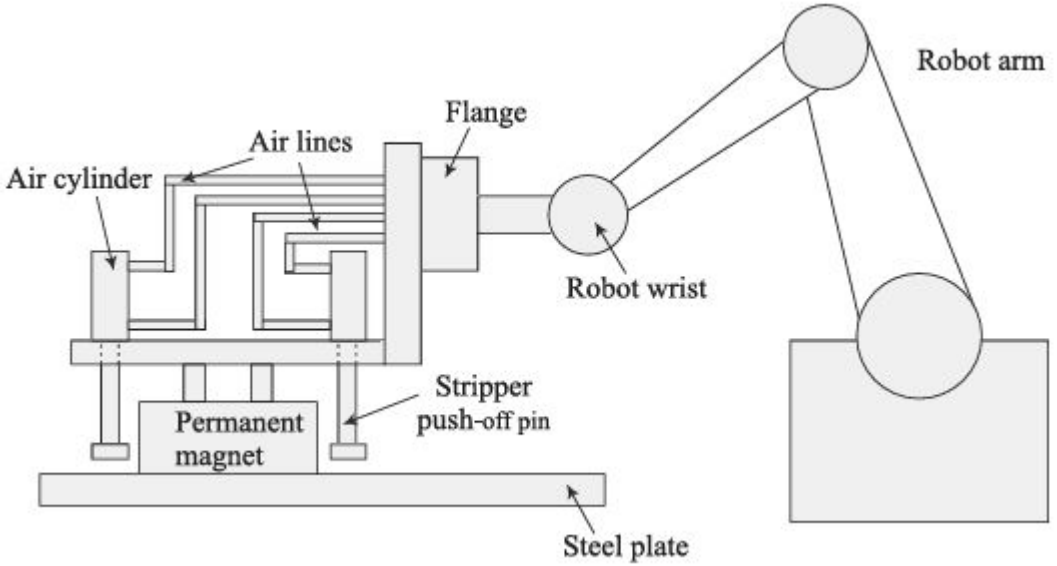


Fig: Motion of a mechanical gripper



Magnetic Grippers

- Unlike mechanical grippers, the principle of a magnetic gripper is based on the magnetic property of a gripper. Hence, they can be used only for ferrous objects. They have the following advantages:
  - Variations in object sizes can be tolerated.
  - Operations are very fast.
  - Require only one surface to hold an object.



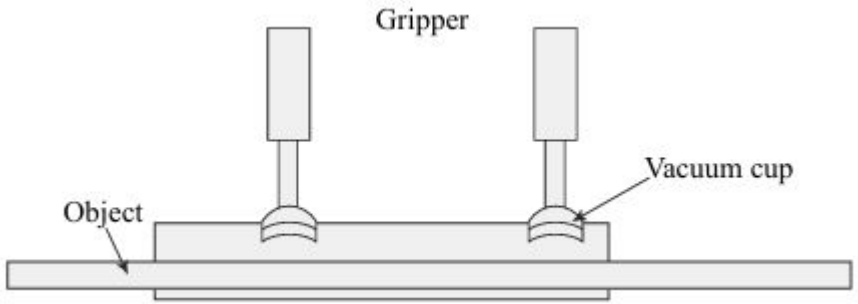
- Magnetic grippers can have either (i) permanent magnets, or (ii) electromagnets. While electromagnetic grippers are easy to control requiring only a dc power source, grippers with permanent magnets do not require any external power source to operate the magnets

Vacuum Grippers

- Vaccum grippers are suitable to handle large flat objects. The material of an object is of no concern with vacuum gripers, except that the object's surface should not have any holes.
- An example of vacuum gripper which uses suction cups made of elastic materials is shown in Fig.
- For a vacuum gripper, lifting capacity can be determined from the negative pressure and the effective area of the cups as

$$f = pA$$

- where f is the force or lift capacity, p is the negative pressure, and A is the total effective area of the suction cups.



Adhesive Grippers

- An adhesive substance used for a grasping action can be used to handle fabrics and other lightweight materials.
- One of the limitations is that the adhesive substance loses its effectiveness with repeated use. Hence, it has to be continuously fed like a mechanical typewriter’s ribbon which needs to be attached to the robot’s wrist.

Hooks, Scoops, and Others

- There exist other types of gripping devices, e.g., hooks, scoops or ladles, inflatable devices, etc., based on the need of item to be handled.



Task Space and Workspace:

Task space

- ✔ The task space is a space in which the robot's task can be naturally expressed.
- ✔ The decision of how to define the task space is driven by the task, independently of the robot.

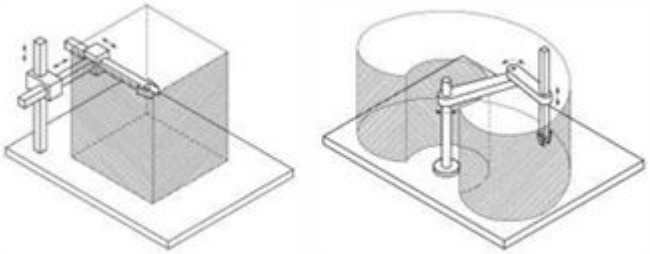


Drawing task space: $\mathbb{R}^2$

Workspace

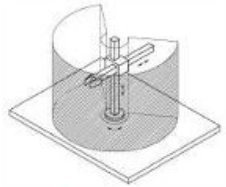
- ✔ Reachable workspace: is a specification of the configurations that the robot end-effector can reach.
- ✔ Dexterous workspace: is a specification of the configurations that the robot end-effector can reach with arbitrary orientation.
- ✔ Robot's workspace depends on: the kinematic configuration, the links' dimension, the joints' range of motion.





Cartesian Manipulator

SCARA Manipulator

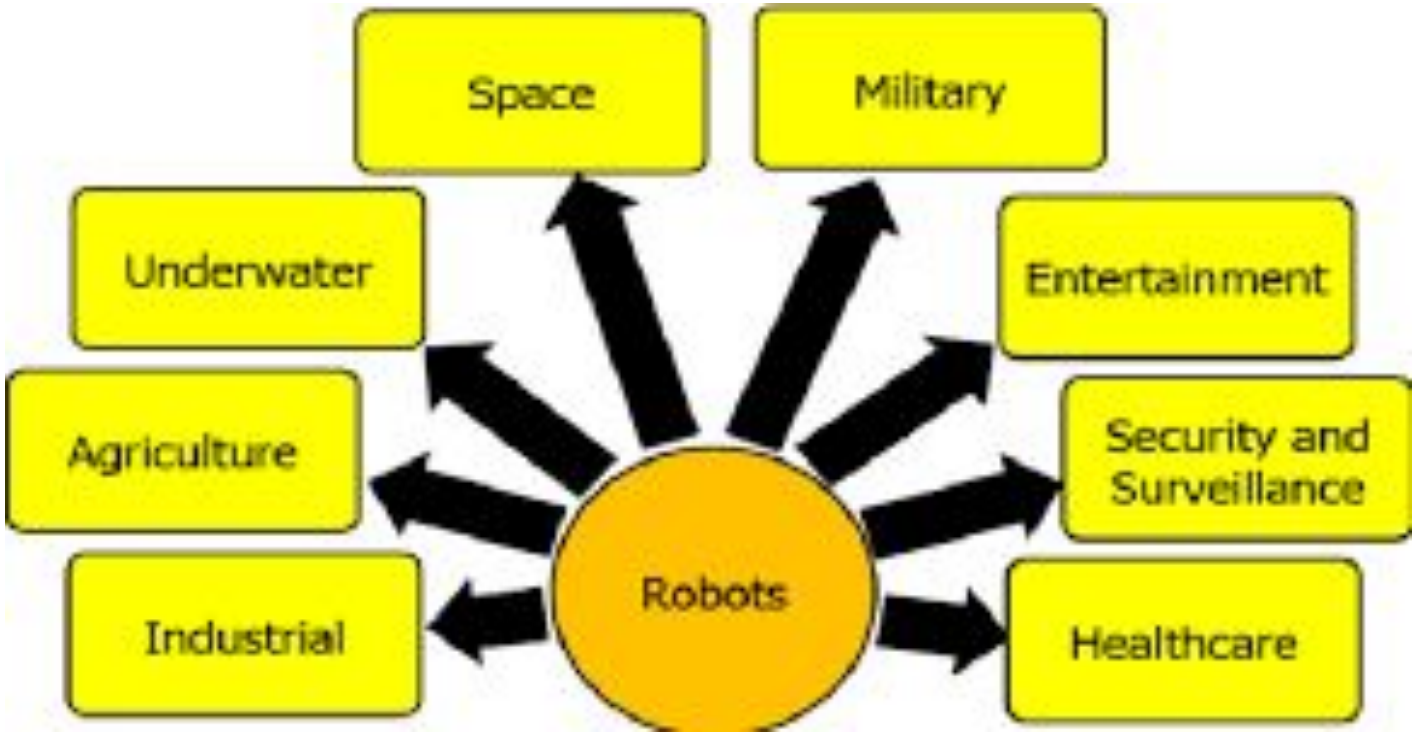


Cylindrical Manipulator

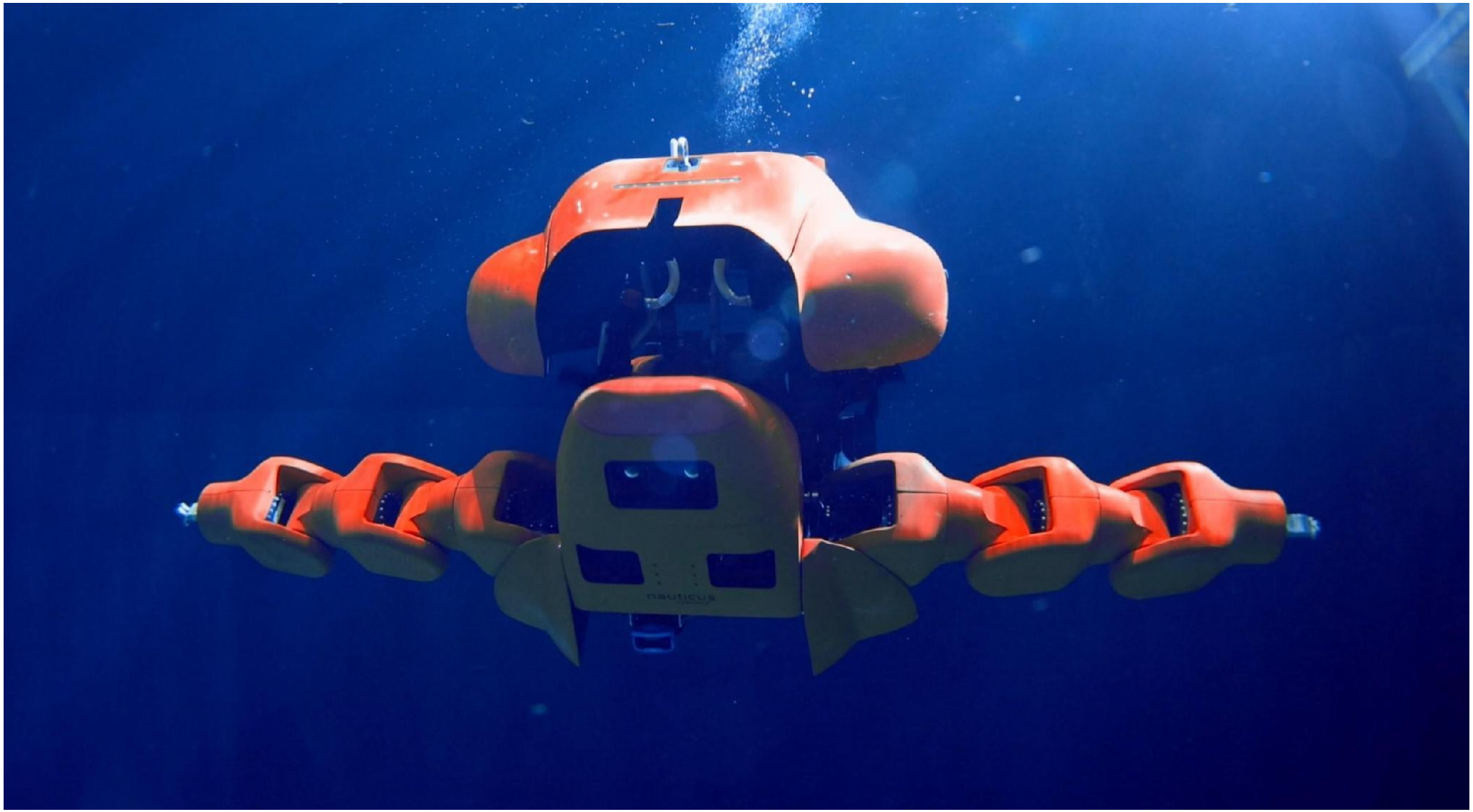
1.10 Applications of Robots

- Machine Loading
- Pick and Place
- Welding
- Painting
- Assembly Operations
- Manufacturing
- Surveillance
- Medical Applications
- Assisting Disabled Individuals
- Hazardous Environments
- Underwater & Space
- Remote Locations

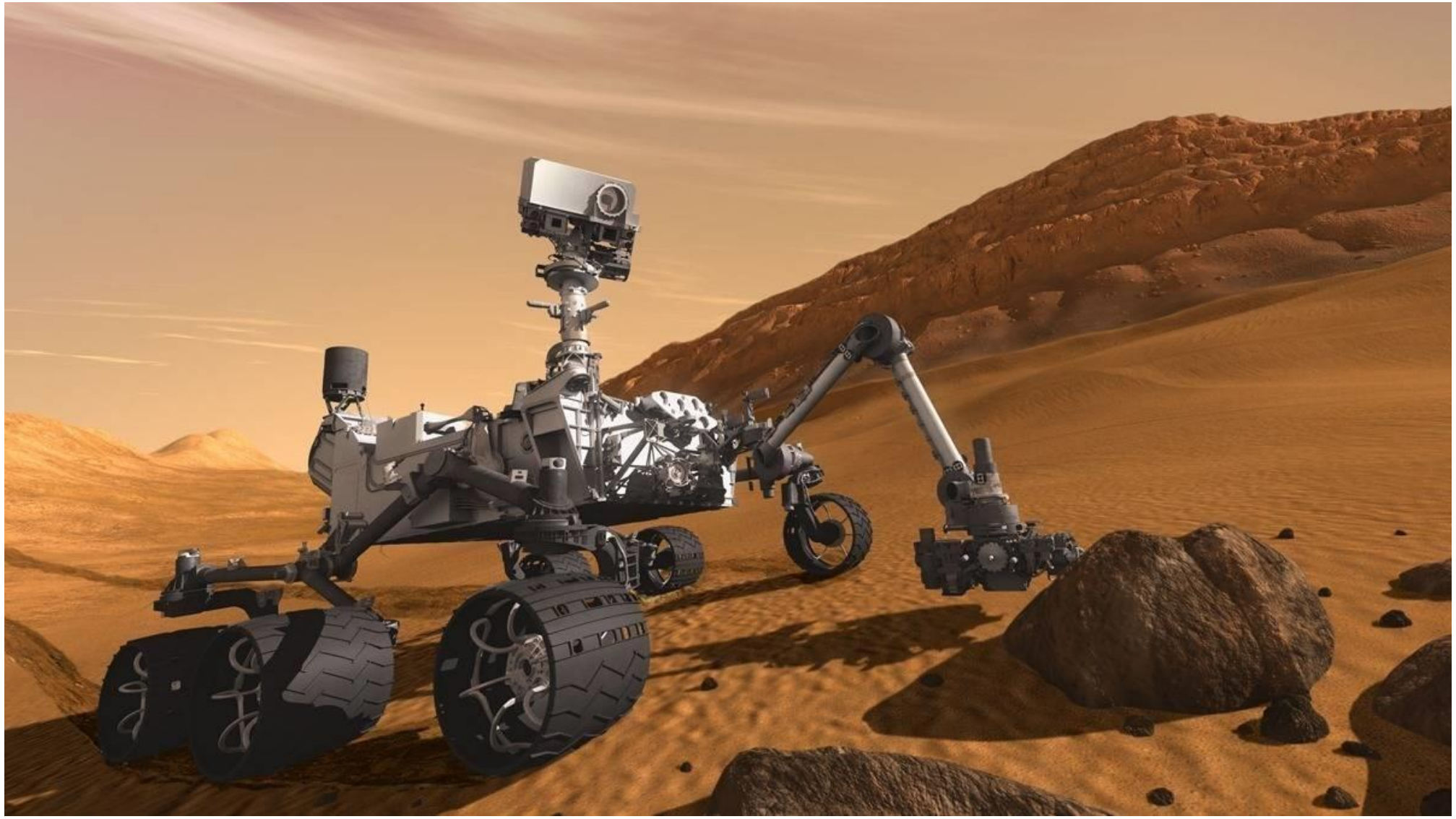
Applications of Robot



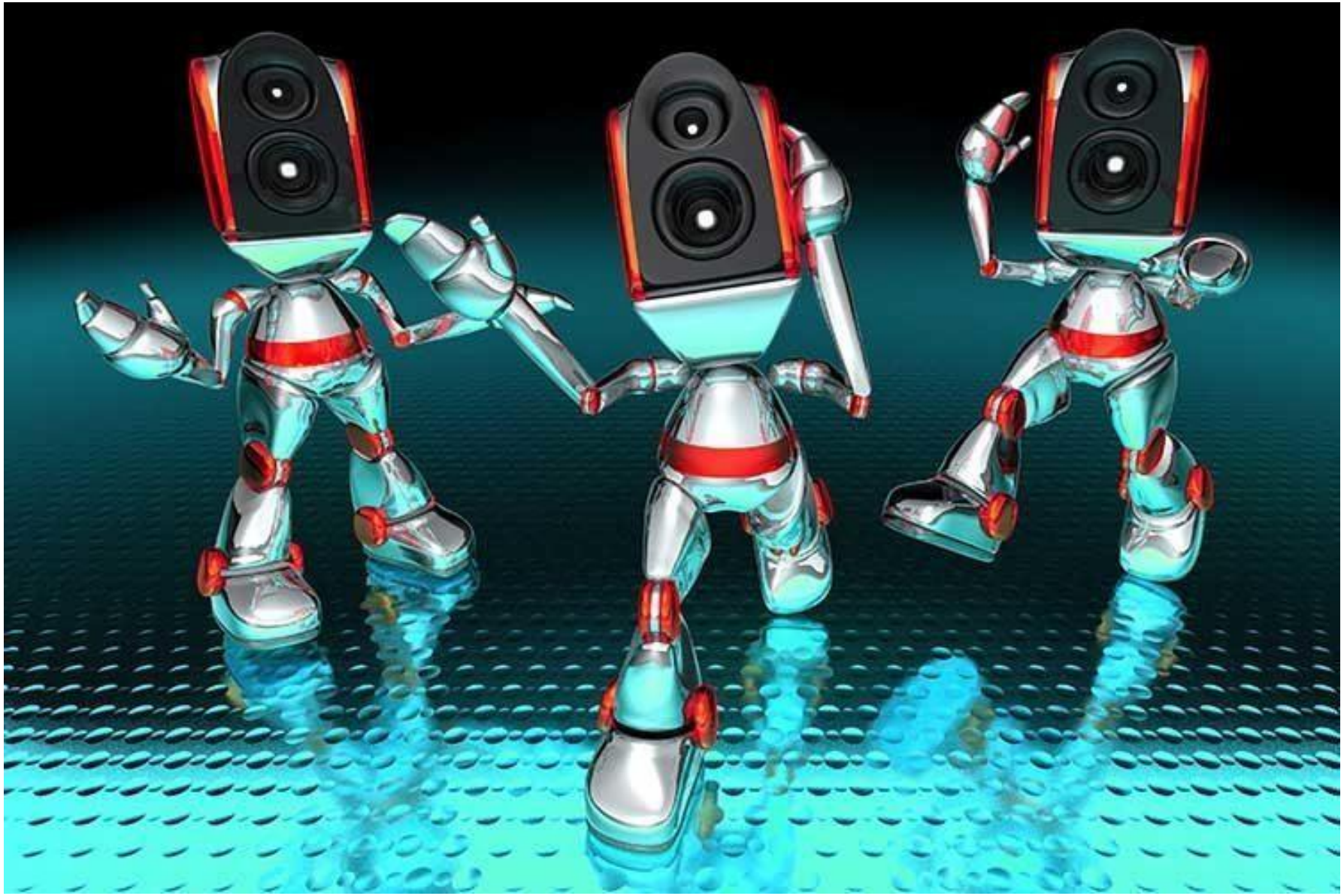


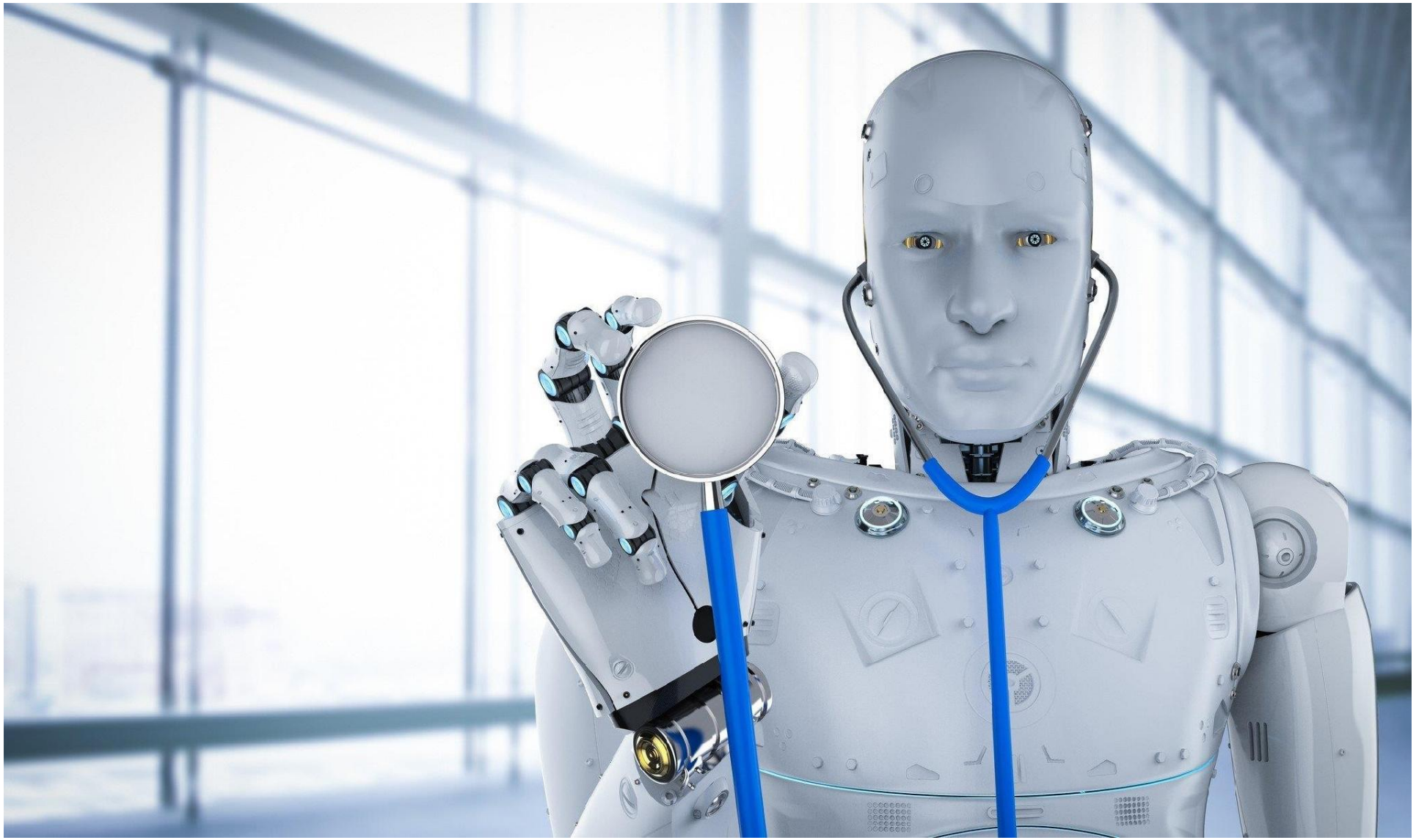













Operating temperature ranges from -25 °C to 45 °C

Applications of Robot

1. Security

   Imagine if all the security guards are robots? Even thieves would be scared! That's why robots are being proposed as security agents as they can protect humans, and they wouldn't be in danger like human security guards would be. Currently, robotics companies are working on pairing robot guards with human security consultants.

2. Agriculture

   Agriculture is the sector that is the basis of human civilization. However, agriculture is also a seasonal sector that is dependent on ideal weather conditions optimal soil, etc. Moreover, there are many repetitive tasks in agriculture that are just a waste of farmer's time and can be performed more suitable by robots. These include seeding, weed control, harvesting, etc. Robots are usually used for harvesting the crops which allow farmers to be more efficient.

Applications of Robot

3. Health Care

Robots have changed healthcare a lot. And all for the better! They can help doctors in performing operations more precisely, be used as prosthetic limbs, provide therapy to patients, etc.

4. Manufacturing

There are many repetitive and common tasks in the manufacturing industry that don't require any usage of the mind like welding, assembly, packing, etc. These tasks can be easily done by robots while leaving the mentally challenging and creative tasks to humans.

5. Military

Robots also have many applications in the military. They can be used as drones to keep surveillance on the enemy, they can also be used as armed systems to attack the opposing forces or as Medicare agents to help friendly forces.

Applications of Robot

5. Military

Robots also have many applications in the military. They can be used as drones to keep surveillance on the enemy, they can also be used as armed systems to attack the opposing forces or as Medicare agents to help friendly forces.

6. Underwater Exploration

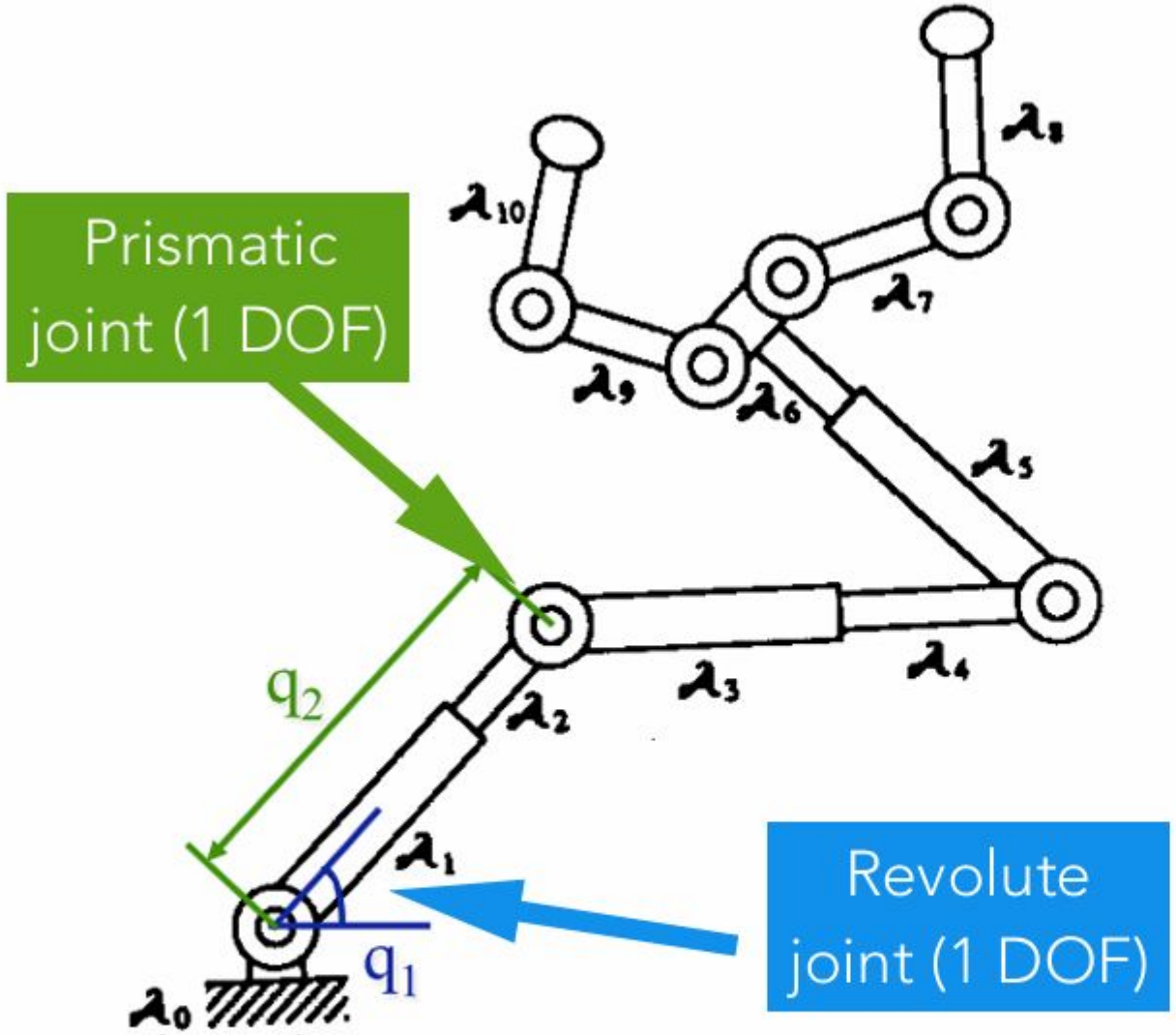
Robots are a great option for exploring places that humans cannot reach easily, like the depths of the ocean! There is a lot of water pressure deep in the ocean which means humans cannot go that down and machines such as submarines can only go to a certain depth as well. These robots are remote-controlled, and they can go into depths of the ocean to collect data and images about the aquatic plant and animal life.

7. Entertainment

Robots are also a big draw in the entertainment industry. While they cannot exactly become actors and actresses. Robots can also be used to do stunt work that is very dangerous for humans but looks pretty cool in an action movie. Theme parks like Disney World are also using autonomous robots to enhance the magical experience of their customers.

Spatial descriptions and Transformations

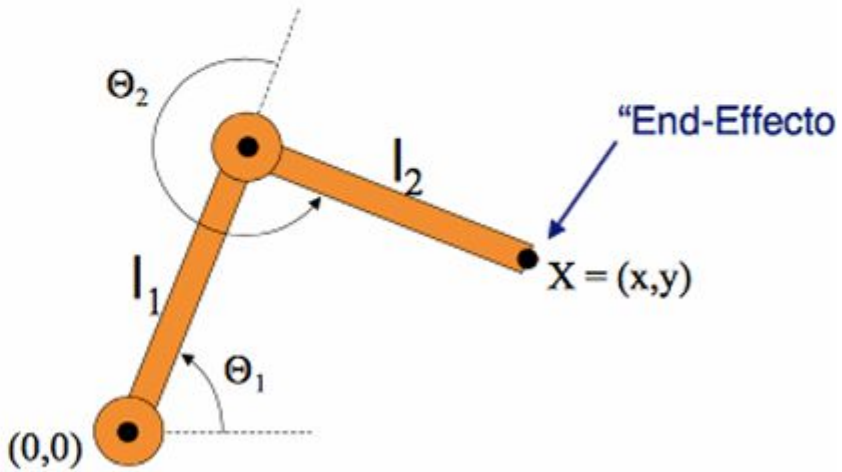




✦ The configuration of a *robot arm* is defined by its joints and their characteristics

$$\vec{q} = (q_1, q_2, \ldots, q_{10})$$

Multiple translational and rotational DOFs, each q is either an angle or a distance



- **Pose of the manipulator:** Configuration of all joints (*n* DoF for *n* revolute joints)
- **Pose of the end-effector:** 6-DoF, can be obtained by the pose of manipulator (reverse doesn't hold! multiple manipulator poses can generate the same ee-pose)

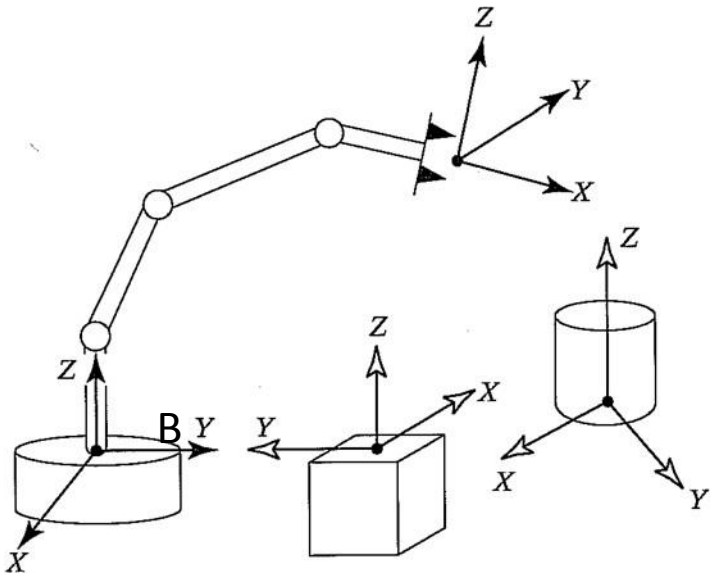
Spatial descriptions

DESCRIPTIONS : POSITIONS,ORIENTATIONS,AND FRAMES

MAPPINGS :
CHANGING DESCRIPTION FROM FRAME TO FRAME

Introduction:

We are constantly concerned with the location of objects in three-dimensional space. These objects are the links of the manipulator, the parts and tools with which it deals, and other objects in the manipulator's environment.





Position and orientation

FIGURE 1.5: Coordinate systems or "frames" are attached to the manipulator and to objects in the environment.

Introduction: cont.

In order to describe the position and orientation of a body in space, we will always attach a coordinate system, or frame, rigidly to the object. We then proceed to describe the position and orientation of this frame with respect to some reference coordinate system. (See Fig. 1.5.)

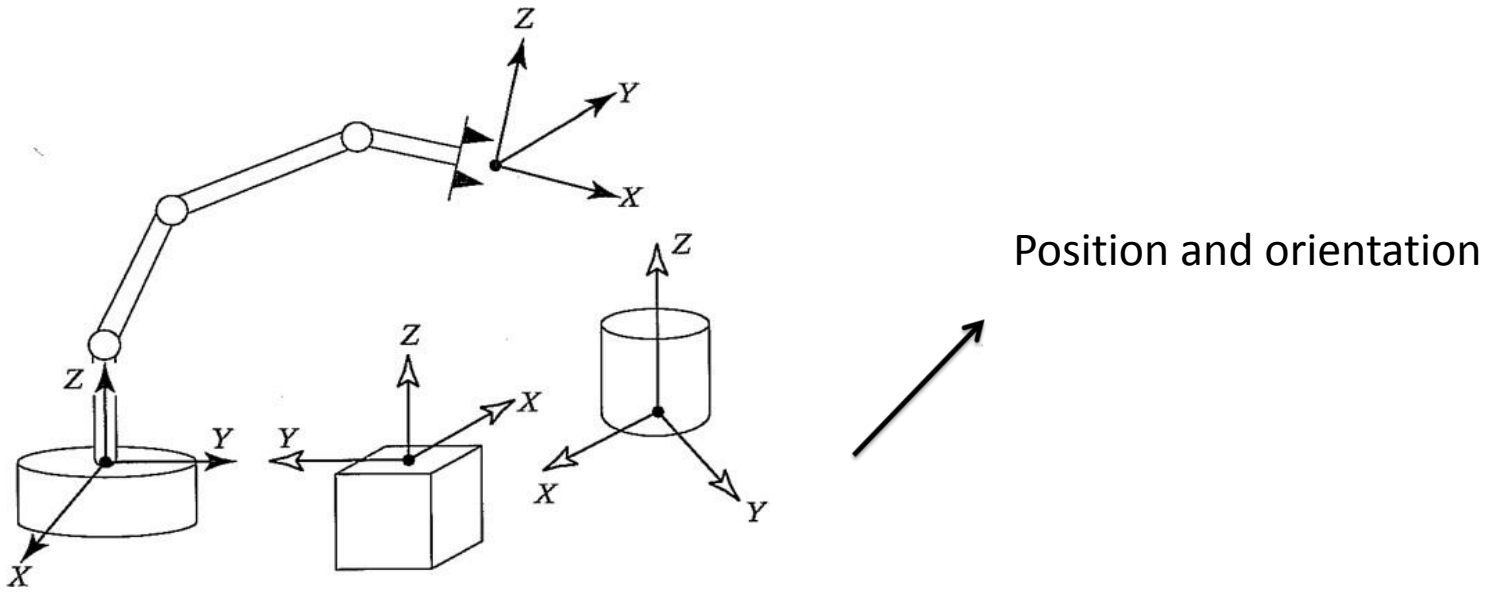
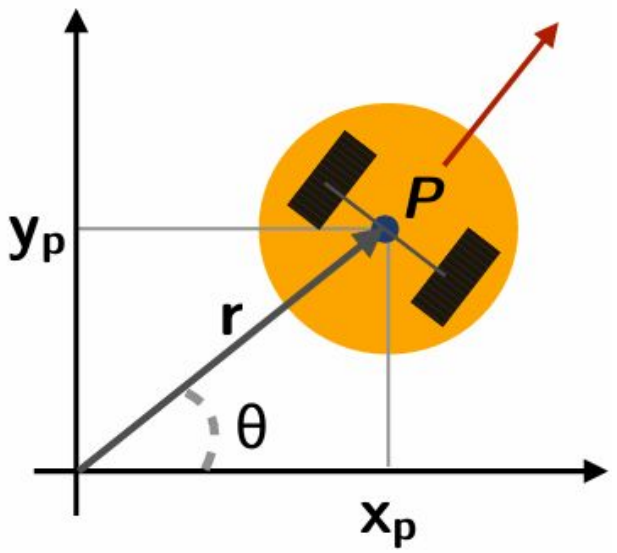


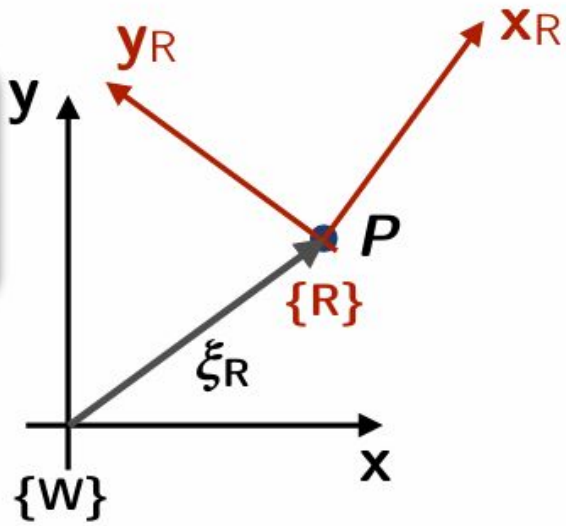
FIGURE 1.5: Coordinate systems or “frames” are attached to the manipulator and to objects in the environment.

Pose

- Define a fixed world reference coordinate frame {W}
- Center a (local) coordinate frame {R} in the robot's reference point *P*, (possibly) oriented according to robot's natural orientation
- A point in space (= chosen reference point *P*) is described by a **coordinate vector *r*** representing the displacement of the point with respect to the *reference coordinate frame* {W} (e.g., using Polar or Cartesian coordinates)



The pose/configuration of the object/robot in {W} is described by the position and orientation of the *(local) coordinate frame* {R} wrt {W}

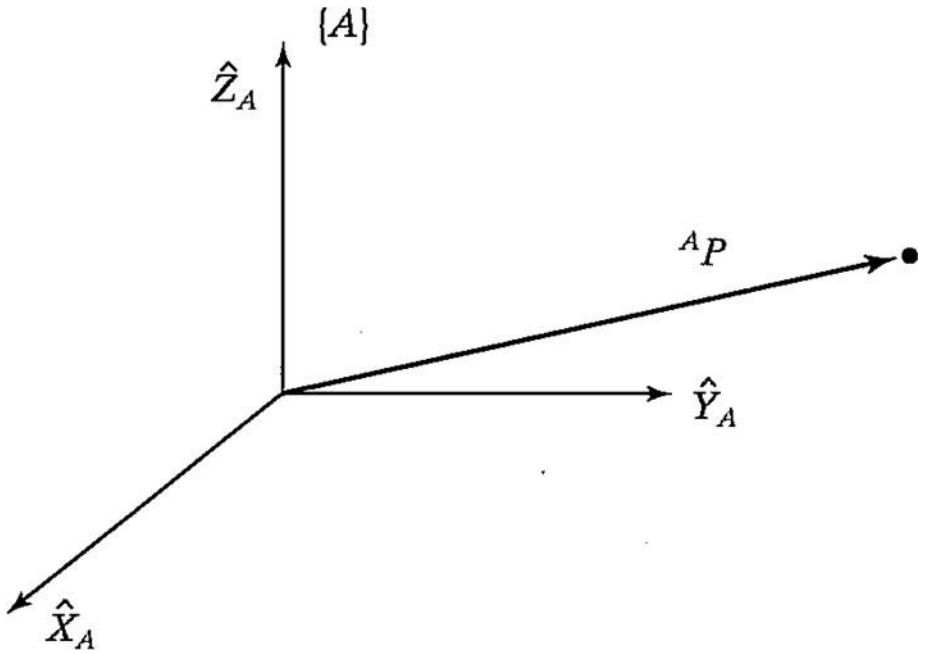


ξ_R is the **relative pose** of the frame/robot with respect to the reference coordinate frame

Description of a position

Once a coordinate system is established, we can locate any point in the universe with a 3 × 1 position vector. Because we will often define many

coordinate systems, the vector will have the name of the coordinate



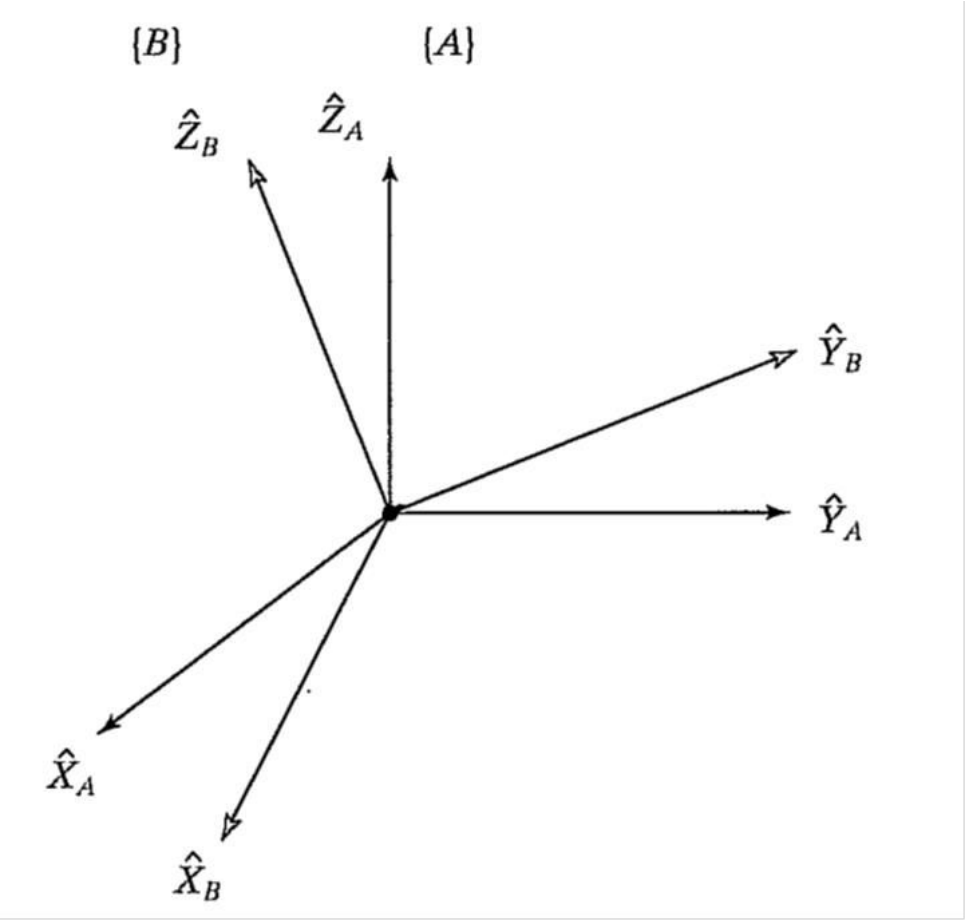
The 'position' of any point, P, on a rigid body in motion with respect to the fixed reference frame can be described by the 3-dimentional Cartesian vector, p, as indicated in Fig. If the coordinates of point P or the components of vector p are, px, py, pz, in the fixed frame A, it is denoted as

$$A_P = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}$$

FIGURE : Vector relative to frame (example).

Description of an orientation

Often, we will find it necessary not only to represent a point in space but also to describe the **orientation** of a body in space .



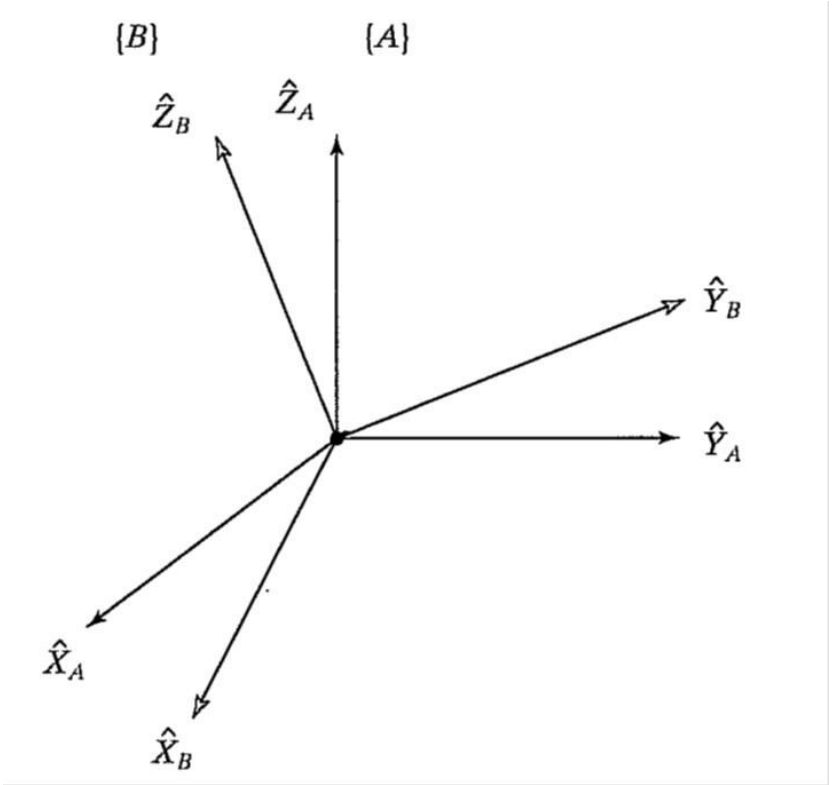
$$ {}^{A}_{B}R = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Rotation matrix : {B} relative to { A}

Rotation matrix : {B} with respect to (w.r.t) A}

We can give expressions for the scalars r_{ij} in (2.2) by nothing that the components of any vector are simply the projections of that vector on to the unit directions of its reference frame. Hence , each component of ${}^{A}_{B}R$ in (2.2) can be written as the dot product of a pair of unit vectors :

$$ {}^{A}_{B}R = \left[{}^{A}\hat{X}_B \quad {}^{A}\hat{Y}_B \quad {}^{A}\hat{Z}_B \right] = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} $$



Note: dot product for vector

Find ${}^{B}_{A}R$ given ${}^{A}_{B}R$?

$$ {}^{A}_{B}R = \begin{bmatrix} {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B} & {}^{A}\hat{Z}_{B} \end{bmatrix} = \begin{bmatrix} {}^{B}\hat{X}^{T}_{A} \\ {}^{B}\hat{Y}^{T}_{A} \\ {}^{B}\hat{Z}^{T}_{A} \end{bmatrix}. $$

Hence, ${}^{B}_{A}R$, the description of frame {A} relative to {B}, is given by the transpose of ${}^{A}R$; that is,

$$ {}^{B}_{A}R = {}^{A}R^{T}. $$

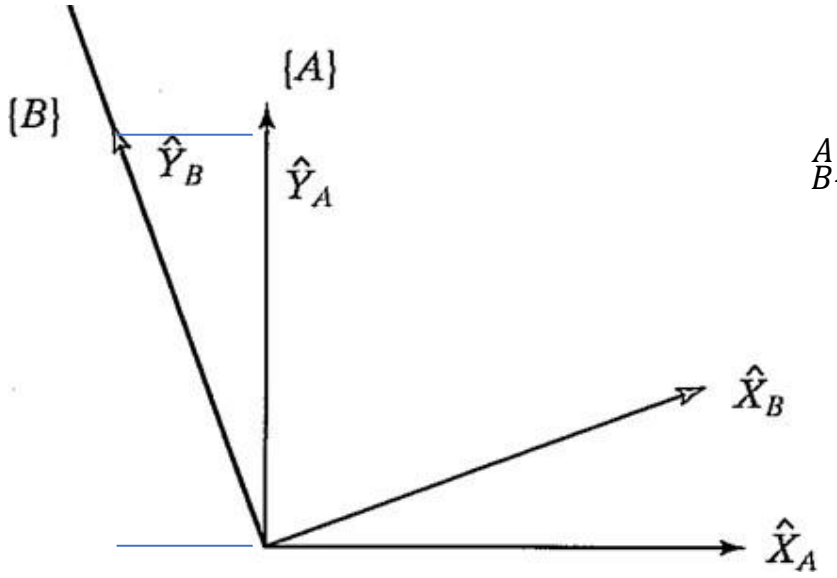
This suggests that the inverse of a rotation matrix is equal to its transpose:

$$ {}^{B}_{A}R = {}^{A}R^{T}_{B} = {}^{A}R^{-1} $$

$$ {}^{A}_{B}R^{T}\ {}^{A}_{B}R = \begin{bmatrix} {}^{A}\hat{X}^{T}_{B} \\ {}^{A}\hat{Y}^{T}_{B} \\ {}^{A}\hat{Z}^{T}_{B} \end{bmatrix} \begin{bmatrix} {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B} & {}^{A}\hat{Z}_{B} \end{bmatrix} = I_3, $$

EXAMPLE 2.1

Figure 2.6 shows a frame {B} that is rotated relative to frame {A} about $_{z\Sigma}$ by θ =30 degrees. Here , $^{z\Sigma}$ is pointing out of the page . Find the rotation matrix {B} w.r.t {A}



$$^{A}_{B}R = R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$^{A}_{B}R = \begin{bmatrix} 0.866 & 0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}$$

$$^{B}_{A}R = {}^{A}_{B}R^{T} = \begin{bmatrix} 0.866 & 0.500 & 0.000 \\ -0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix}$$

FIGURE 2.6: {B} rotated 30 degrees about z Σ

.

APPENDIX A

Formulas for rotation about the principle axes by :

$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix},$$

$$R_Y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix},$$

$$R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Mappings involving general frames

Very often, we know the description of a vector with respect to some frame {B},and we would like to know its description with respect to another frame,{A}.

We now consider the general case of mapping. Here, the origin of frame {B} is not coincident with that of frame {A} but has a general vector offset. The vector that locates {B}'s origin is called ${}^{A}P_{BORG}$. Also {B} is rotated with respect to {A}, as described by ${}^{A}_{B}R$.

Given ${}^{B}P$, we wish to compute ${}^{A}P$ as in Fig. 2.7.

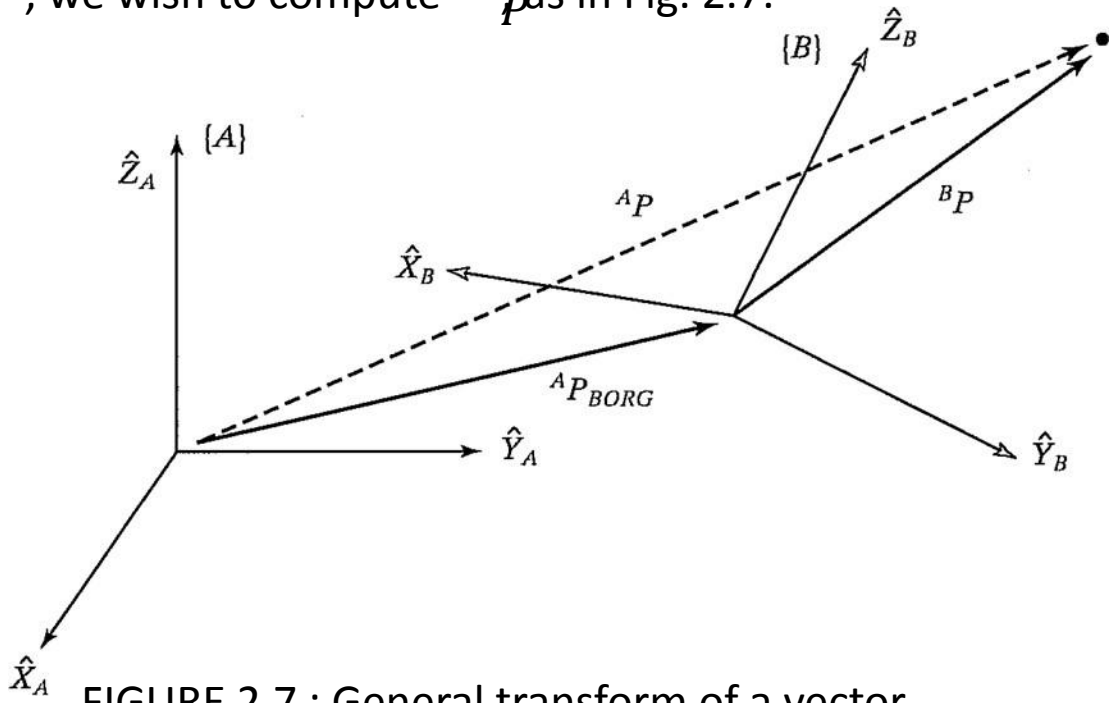
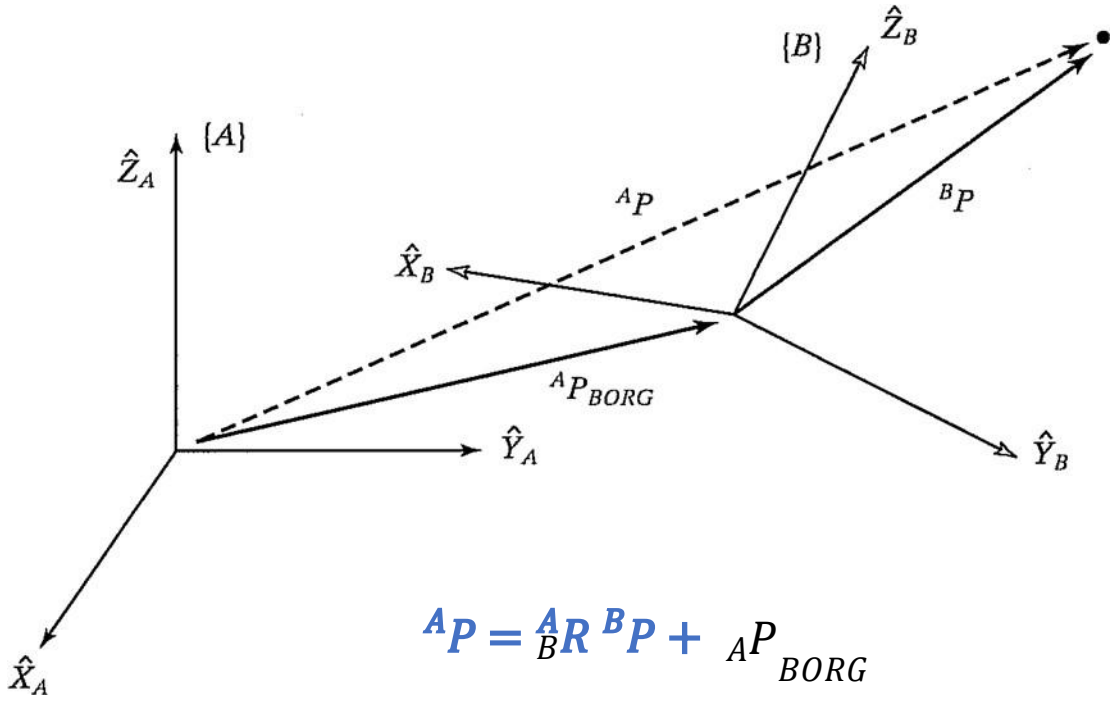


FIGURE 2.7 : General transform of a vector .



$$ {}^AP = {}^A_BR\,{}^BP + {}_AP_{BORG} $$

Homogeneous transform

$$ {}^AP = {}^A_BT\,{}^BP $$

$$ \begin{bmatrix} {}^AP \\ 1 \end{bmatrix} = {}^A_BR \begin{bmatrix} {}^BP \\ 1 \end{bmatrix} $$

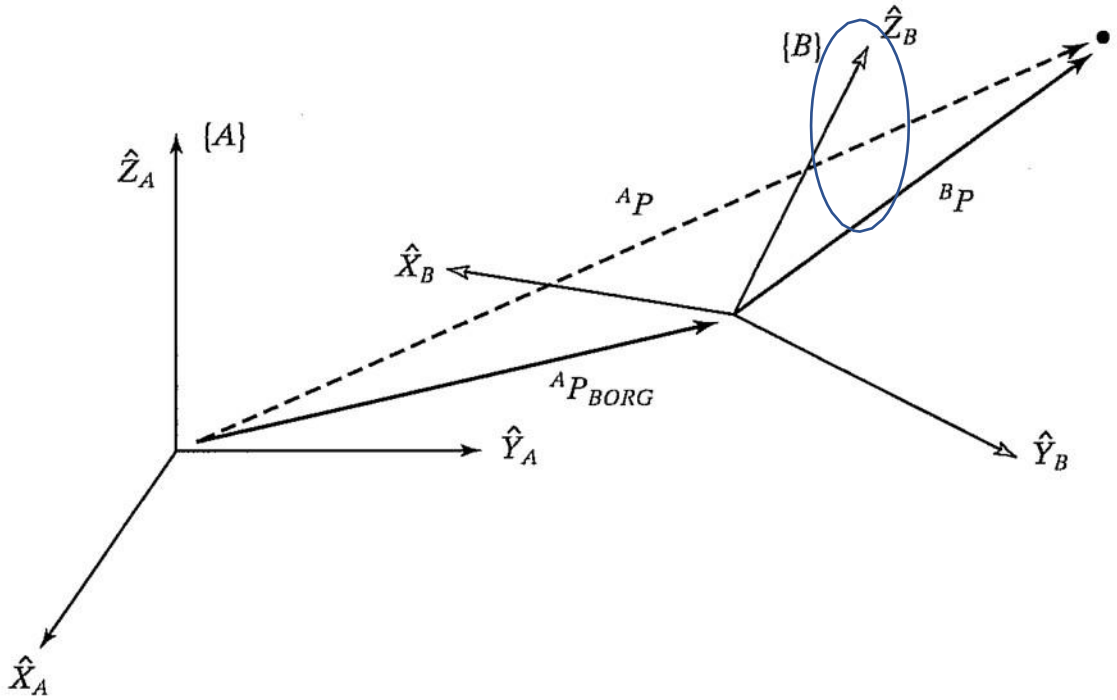
$$ {}^A_BT = \begin{bmatrix} {}^A_BR & {}_AP_{BORG} \\ 0\ 0\ 0 & 1 \end{bmatrix} $$

EXAMPLE 2.2

Figure shows a frame {B}, which is rotated relative to frame {A} about

by 30 degree, and translated 10 units in Z_A, and translated 5 units in Y_A. Find ${}^{A}P$, where ${}^{B}P = [3.0\ 7.0\ 0.0]^T$.

$$ {}^{A}_{B}R = R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ 0.000 & \cos\theta & 0 \\ \sin\theta & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.500 & \\ 0.500 & 0.866 & \\ 0.000 & & \end{bmatrix} $$



EXAMPLE 2.5

Figure 2.13 shows a frame {B} that is rotated relative to frame {A} about Z by 30 degrees and translated four units in X_A and three units in Y_A. Thus, we have a description of ${}^{A}_{B}T$.Find ${}^{B}_{A}T$?

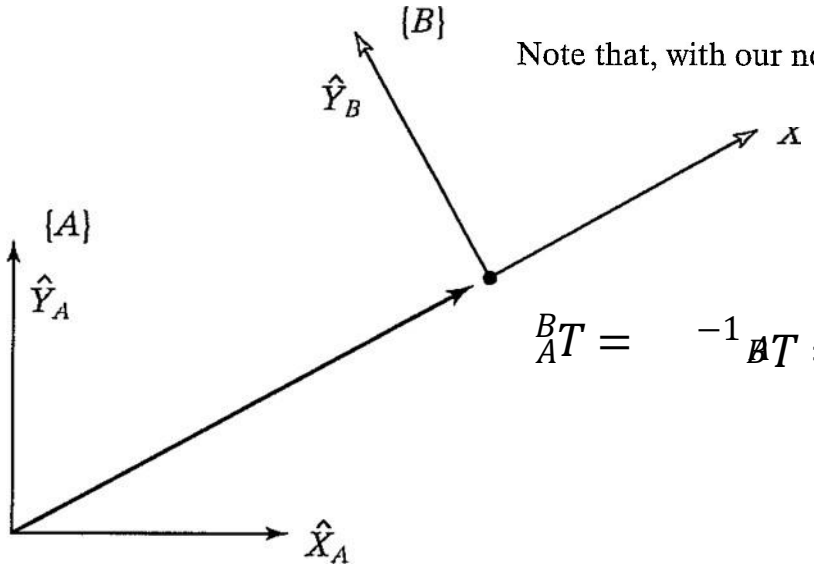
$$ {}^{A}_{B}T = \begin{bmatrix} 0.866 & -0.5 & 0 & 4 \\ 0.5 & 0.866 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad (2.68) $$

Using (2.42) and (2.44), we can write the form of ${}^{B}_{A}T$ as

$$ {}^{B}_{A}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\,{}^{A}P_{BORG} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]. $$

Note that, with our notation,

$$ {}^{B}_{A}T = {}^{A}_{B}T^{-1}. $$

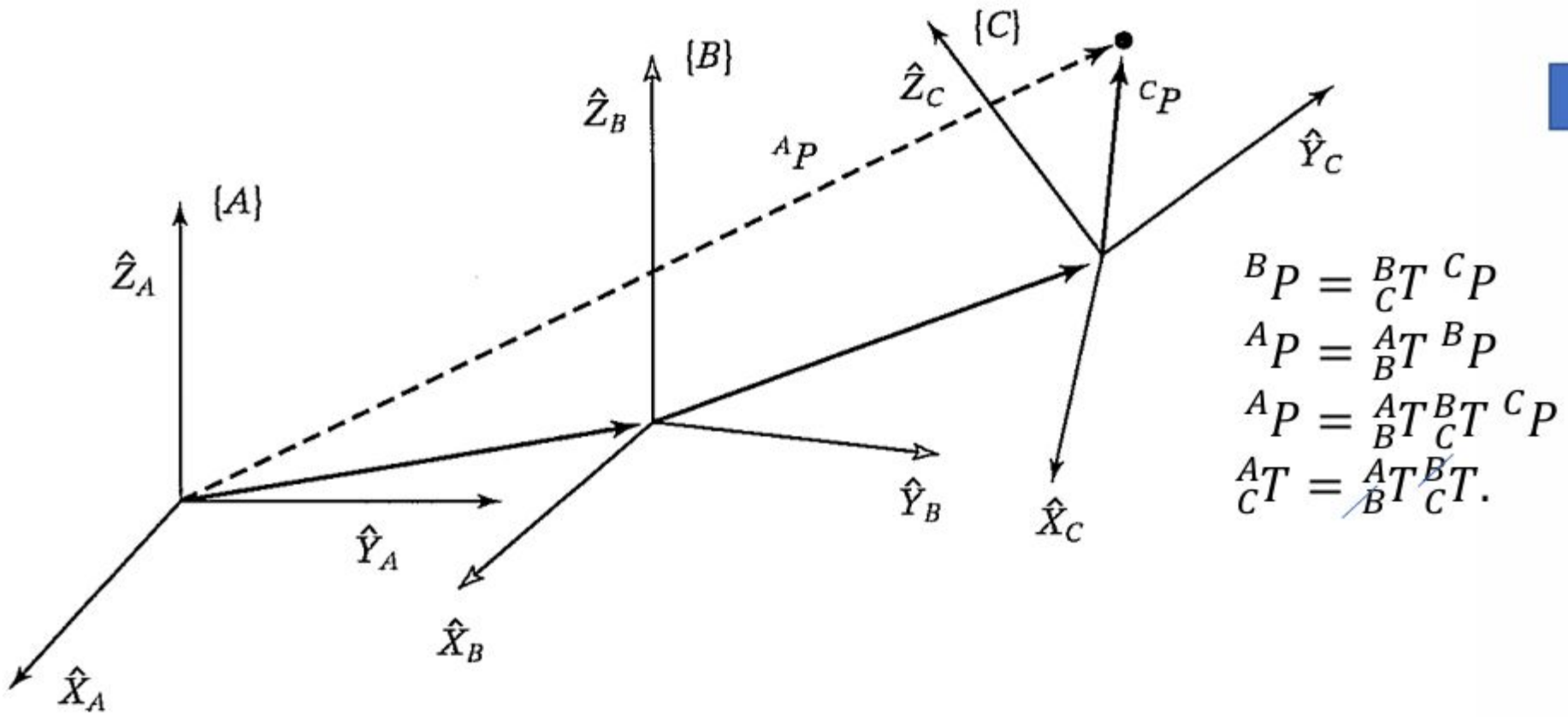


$$ {}^{B}_{A}T = {}^{-1}_{B}T = \begin{bmatrix} 0.866 & 0.500 & 0 & -4.964 \\ -0.5 & 0.866 & 0 & -0.598 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

FIGURE 2.13 : {B} relative to {A} .

Compound transformations

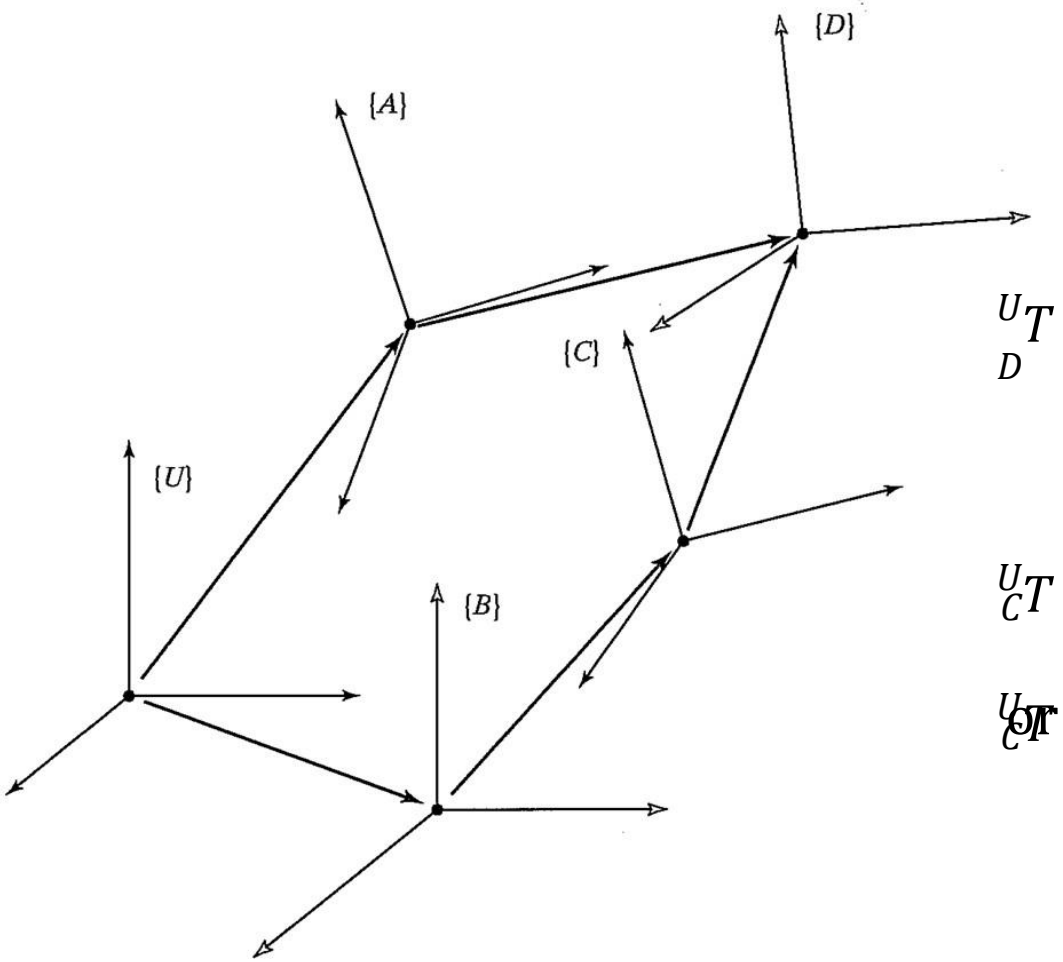
we have ${}^{C}P$ and wish to find ${}^{A}P$.



$$
\begin{aligned}
{}^{B}P &= {}^{B}_{C}T\ {}^{C}P \\
{}^{A}P &= {}^{A}_{B}T\ {}^{B}P \\
{}^{A}P &= {}^{A}_{B}T\,{}^{B}_{C}T\ {}^{C}P \\
{}^{A}_{C}T &= {}^{A}_{B}T\,{}^{B}_{C}T.
\end{aligned}
$$

FIGURE : Compound frames: Each is known relative to previous one .

2.7 TRANSFORM EQUATIONS



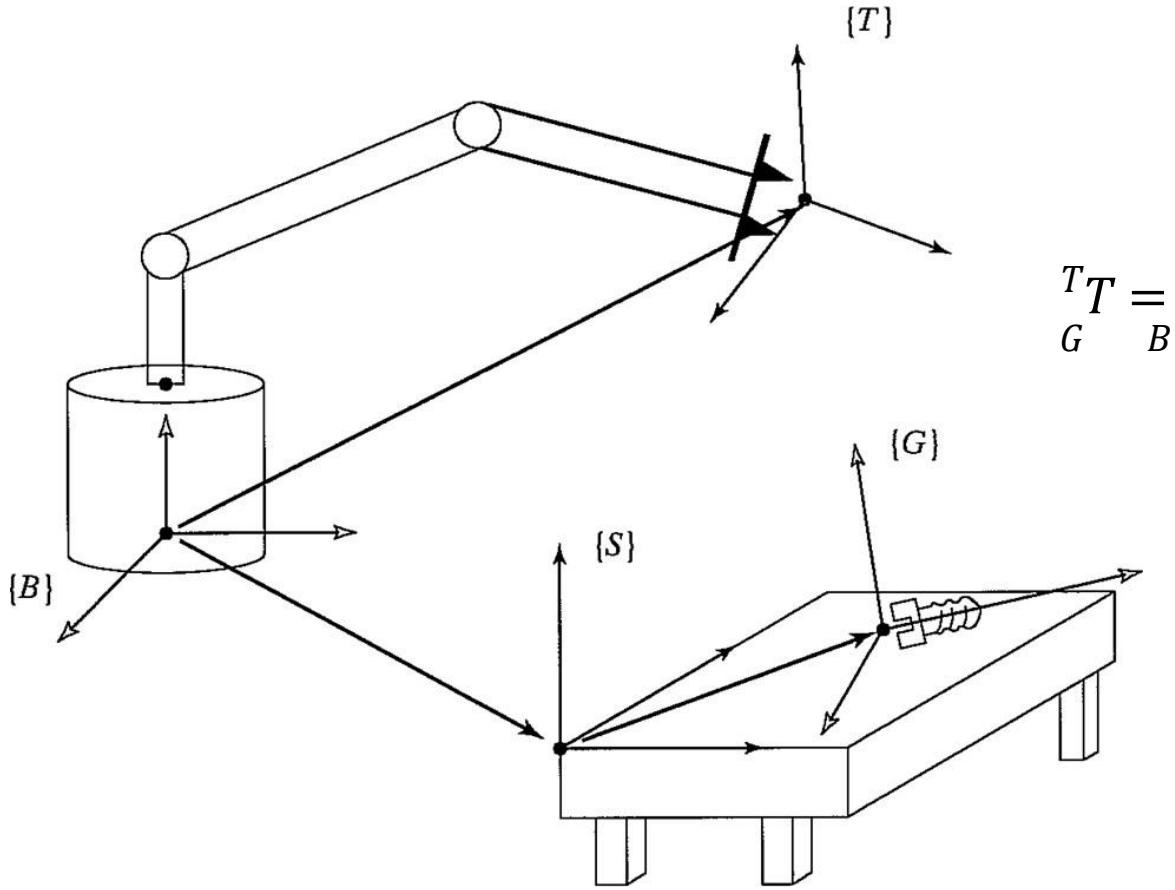
$$^{U}_{D}T = {}^{U}_{A}T\,{}^{A}_{D}T, \text{ or } = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T.$$

$$^{U}_{C}T = {}^{U}_{A}T\,{}^{A}_{D}T\,{}^{D}_{C}T = {}^{U}_{A}T\,{}^{A}_{D}T\,{}^{C}_{D}T^{-1}$$

$$\text{Or } {}^{U}_{C}T = {}^{U}_{B}T\,{}^{B}_{C}T.$$

FIGURE 2.14: Set of transforms forming a loop .

2.7 TRANSFORM EQUATIONS



$$ {}^{T}_{G}T = {}^{T}_{B}T\,{}^{B}_{S}T\,{}^{S}_{G}T = {}^{B}_{T}T^{-1}\,{}^{B}_{S}T\,{}^{S}_{G}T. $$

FIGURE 2.16 : Manipulator reaching for a bolt .

MORE ON REPRESENTATION OF ORIANTATION

Rotation matrix determinant is +1

$$R = [\hat{X}\ \hat{Y}\ \hat{Z}] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\begin{aligned} |X| &= 1, \\ |\hat{Y}| &= 1, \\ |\hat{Z}| &= 1, \\ \hat{X} \,.\, \hat{Y} &= 0, \\ \hat{X} \,.\, \hat{Z} &= 0, \\ \hat{Y} \,.\, \hat{Z} &= 0. \end{aligned}$$

- Clearly, the nine elements of a rotation matrix are not all independent .
- In fact, given a rotation matrix, R , it is easy to write down the six dependencies between the elements.
- Therefore, rotation matrix can be specified by just three parameters.

2.8 MORE ON REPRESENTATION OF ORIANTATION

- Rotation matrices are useful as operators (computer). Their matrix form is such that, when multiplied by a vector , they perform the rotation operation.

- Human operator at a computer terminal who wishes to type in the specification of the desired orientation of a robot's hand would have a hard time inputting a nine-element matrix with orthonormal columns. A representation that requires only three numbers would be simpler .